Chapter 8

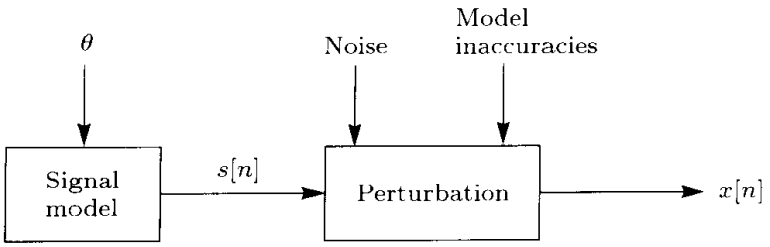
Least Squares

8.1 Introduction

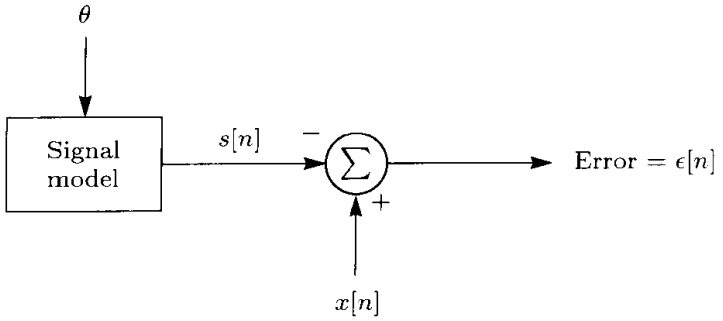
In previous chapters we have attempted to find an optimal or nearly optimal (for large data records) estimator by considering the class of unbiased estimators and determining the one exhibiting minimum variance, the so-called MVU estimator. We now depart from this philosophy to investigate a class of estimators that in general have no optimality properties associated with them but make good sense for many problems of interest. This is the method of least squares and dates back to 1795 when Gauss used the method to study planetary motions. A salient feature of the method is that no probabilistic assumptions are made about the data, only a signal model is assumed. The advantage, then, is its broader range of possible applications. On the negative side, no claims about optimality can be made, and furthermore, the statistical performance cannot be assessed without some specific assumptions about the probabilistic structure of the data. Nonetheless, the least squares estimator is widely used in practice due to its ease of implementation, amounting to the minimization of a least squares error criterion.

8.2 Summary

The least squares approach to parameter estimation chooses θ to minimize (8.1), where the signal depends on θ. Linear versus nonlinear least squares problems are described in Section 8.3. The general linear least squares problem which minimizes (8.9) leads to the least squares estimator of (8.10) and the minimum least squares error of (8.11)–(8.13). A weighted least squares error criterion of (8.14) results in the estimator of (8.16) and the minimum least squares error of (8.17). A geometrical interpretation of least squares is described in Section 8.5 and leads to the important orthogonality principle. When the dimensionality of the vector parameter is not known, an order-recursive least squares approach can be useful. It computes the least squares estimator recursively as the number of unknown parameters increases. It is summarized by (8.28)–(8.31). If it is desired to update in time the least squares estimator as more data become



(a) Data model



(b) Least squares error

Figure 8.1 Least squares approach

available, then a sequential approach can be employed. It determines the least squares estimator based on the estimate at the previous time and the new data. The equations (8.46)–(8.48) summarize the calculations required. At times the parameter vector is constrained, as in (8.50). In such a case the constrained least squares estimator is given by (8.52). Nonlinear least squares is discussed in Section 8.9. Some methods for converting the problem to a linear one are described, followed by iterative minimization approaches if this is not possible. The two methods that are generally used are the Newton-Raphson iteration of (8.61) and the Gauss-Newton iteration of (8.62).

8.3 The Least Squares Approach

Our focus in determining a good estimator has been to find one that was unbiased and had minimum variance. In choosing the variance as our measure of goodness we implicitly sought to minimize the discrepancy (on the average) between our estimate and the true parameter value. In the least squares (LS) approach we attempt to minimize the squared difference between the given data $x[n]$ and the assumed signal or noiseless data. This is illustrated in Figure 8.1. The signal is generated by some model which in turn depends upon our unknown parameter θ. The signal $s[n]$ is purely deterministic. Due to observation noise or model inaccuracies we observe a perturbed version of $s[n]$, which we denote by $x[n]$. The least squares estimator (LSE) of θ chooses the value that

makes $s[n]$ closest to the observed data $x[n]$. Closeness is measured by the LS error criterion

$$J(\theta) = \sum_{n=0}^{N-1} (x[n] - s[n])^2 \tag{8.1}$$

where the observation interval is assumed to be $n = 0, 1, \ldots, N-1$, and the dependence of J on θ is via $s[n]$. The value of θ that minimizes $J(\theta)$ is the LSE. *Note that no probabilistic assumptions have been made about the data $x[n]$.* The method is equally valid for Gaussian as well as non-Gaussian noise. Of course, the *performance* of the LSE will undoubtedly depend upon the properties of the corrupting noise as well as any modeling errors. LSEs are usually applied in situations where a precise statistical characterization of the data is unknown or where an optimal estimator cannot be found or may be too complicated to apply in practice.

Example 8.1 - DC Level Signal

Assume that the signal model in Figure 8.1 is $s[n] = A$ and we observe $x[n]$ for $n = 0, 1, \ldots, N-1$. Then, according to the LS approach, we can estimate A by minimizing (8.1) or

$$J(A) = \sum_{n=0}^{N-1} (x[n] - A)^2.$$

Differentiating with respect to A and setting the result equal to zero produces

$$\begin{aligned} \hat{A} &= \frac{1}{N} \sum_{n=0}^{N-1} x[n] \\ &= \bar{x} \end{aligned}$$

or the sample mean estimator. Our familiar estimator, however, cannot be claimed to be optimal in the MVU sense but only in that it minimizes the LS error. We know, however, from our previous discussions that if $x[n] = A + w[n]$, where $w[n]$ is zero mean WGN, then the LSE will also be the MVU estimator, but otherwise not. To underscore the potential difficulties, consider what would happen if the noise were not zero mean. Then, the sample mean estimator would actually be an estimator of $A + E(w[n])$ since $w[n]$ could be written as

$$w[n] = E(w[n]) + w'[n]$$

where $w'[n]$ is zero mean noise. The data are more appropriately described by

$$x[n] = A + E(w[n]) + w'[n].$$

It should be clear that in using this approach, it must be assumed that the observed data are composed of a deterministic signal and *zero mean* noise. If this is the case, the error $\epsilon[n] = x[n] - s[n]$ will tend to be zero on the average for the correct choice of the signal parameters. The minimization of (8.1) is then a reasonable approach. The reader might also consider what would happen if the assumed DC level signal model

were incorrect, as for instance if $x[n] = A+Bn+w[n]$ described the data. This modeling error would also cause the LSE to be biased. $\diamond$

Example 8.2 - Sinusoidal Frequency Estimation

Consider the signal model

$$s[n] = \cos 2\pi f_0 n$$

in which the frequency f_0 is to be estimated. The LSE is found by minimizing

$$J(f_0) = \sum_{n=0}^{N-1} (x[n] - \cos 2\pi f_0 n)^2.$$

In contrast to the DC level signal for which the minimum is easily found, here the LS error is highly nonlinear in f_0. The minimization cannot be done in closed form. Since the error criterion is a quadratic function of the signal, a signal that is *linear* in the unknown parameter yields a quadratic function for J, as in the previous example. The minimization is then easily carried out. A signal model that is *linear in the unknown parameter* is said to generate a *linear least squares* problem. Otherwise, as in this example, the problem is a *nonlinear least squares* problem. Nonlinear LS problems are solved via grid searches or iterative minimization methods as described in Section 8.9. It should be noted that the signal itself need not be linear but only in the unknown parameter, as the next example illustrates. $\diamond$

Example 8.3 - Sinusoidal Amplitude Estimation

If the signal is $s[n] = A\cos 2\pi f_0 n$, where f_0 *is known* and A is to be estimated, then the LSE minimizes

$$J(A) = \sum_{n=0}^{N-1} (x[n] - A\cos 2\pi f_0 n)^2$$

over A. This is easily accomplished by differentiation since $J(A)$ is quadratic in A. This linear LS problem is clearly very desirable from a practical viewpoint. If, however, A were known and the frequency were to be estimated, the problem would be equivalent to that in Example 8.2, that is to say, it would be a nonlinear LS problem. A final possibility, in the vector parameter case, is that both A and f_0 might need to be estimated. Then, the error criterion

$$J(A, f_0) = \sum_{n=0}^{N-1} (x[n] - A\cos 2\pi f_0 n)^2$$

is quadratic in A but nonquadratic in f_0. The net result is that J can be minimized in closed form with respect to A for a given f_0, reducing the minimization of J to one

over f_0 only. This type of problem, in which the signal is linear in some parameters but nonlinear in others, is termed a *separable least squares* problem. In Section 8.9 we will discuss this further. ◇

8.4 Linear Least Squares

In applying the linear LS approach for a scalar parameter we must assume that

$$s[n] = \theta h[n] \tag{8.2}$$

where $h[n]$ is a known sequence. (The reader may want to refer back to Chapter 6 on the BLUE for a comparison to similar signal models.) The LS error criterion becomes

$$J(\theta) = \sum_{n=0}^{N-1} (x[n] - \theta h[n])^2. \tag{8.3}$$

A minimization is readily shown to produce the LSE

$$\hat{\theta} = \frac{\sum_{n=0}^{N-1} x[n]h[n]}{\sum_{n=0}^{N-1} h^2[n]}. \tag{8.4}$$

The minimum LS error, obtained by substituting (8.4) into (8.3), is

$$\begin{aligned} J_{\min} = J(\hat{\theta}) &= \sum_{n=0}^{N-1} (x[n] - \hat{\theta}h[n])(x[n] - \hat{\theta}h[n]) \\ &= \sum_{n=0}^{N-1} x[n](x[n] - \hat{\theta}h[n]) - \hat{\theta} \underbrace{\sum_{n=0}^{N-1} h[n](x[n] - \hat{\theta}h[n])}_{S} \\ &= \sum_{n=0}^{N-1} x^2[n] - \hat{\theta} \sum_{n=0}^{N-1} x[n]h[n]. \end{aligned} \tag{8.5}$$

The last step follows because the sum S is zero (substitute $\hat{\theta}$ to verify). Alternatively, by using (8.4) we can rewrite $J_{\min}$ as

$$J_{\min} = \sum_{n=0}^{N-1} x^2[n] - \frac{\left(\sum_{n=0}^{N-1} x[n]h[n]\right)^2}{\sum_{n=0}^{N-1} h^2[n]}. \tag{8.6}$$

The minimum LS error is the original energy of the data or $\sum_{n=0}^{N-1} x^2[n]$ less that due to the signal fitting. For Example 8.1 in which $\theta = A$ we have $h[n] = 1$, so that from (8.4) $\hat{A} = \bar{x}$ and from (8.5)

$$J_{\min} = \sum_{n=0}^{N-1} x^2[n] - N\bar{x}^2.$$

If the data were noiseless so that $x[n] = A$, then $J_{\min} = 0$ or we would have a perfect LS fit to the data. On the other hand, if $x[n] = A + w[n]$, where $E(w^2[n]) \gg A^2$, then $\sum_{n=0}^{N-1} x^2[n]/N \gg \bar{x}^2$. The minimum LS error would then be

$$J_{\min} \approx \sum_{n=0}^{N-1} x^2[n]$$

or not much different than the original error. It can be shown (see Problem 8.2) that the minimum LS error is always between these two extremes or

$$0 \leq J_{\min} \leq \sum_{n=0}^{N-1} x^2[n]. \tag{8.7}$$

The extension of these results to a vector parameter $\boldsymbol{\theta}$ of dimension $p \times 1$ is straightforward and of great practical utility. For the signal $\mathbf{s} = [s[0]\, s[1] \ldots s[N-1]]^T$ to be linear in the unknown parameters, we assume, using matrix notation,

$$\mathbf{s} = \mathbf{H}\boldsymbol{\theta} \tag{8.8}$$

where $\mathbf{H}$ is a known $N \times p$ matrix ($N > p$) of full rank p. The matrix $\mathbf{H}$ is referred to as the *observation matrix*. This is, of course, the linear model, albeit without the usual noise PDF assumption. Many examples of signals satisfying this model can be found in Chapter 4. The LSE is found by minimizing

$$\begin{aligned} J(\boldsymbol{\theta}) &= \sum_{n=0}^{N-1} (x[n] - s[n])^2 \\ &= (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}). \end{aligned} \tag{8.9}$$

This is easily accomplished (since J is a quadratic function of $\boldsymbol{\theta}$) by using (4.3). Since

$$\begin{aligned} J(\boldsymbol{\theta}) &= \mathbf{x}^T\mathbf{x} - \mathbf{x}^T\mathbf{H}\boldsymbol{\theta} - \boldsymbol{\theta}^T\mathbf{H}^T\mathbf{x} + \boldsymbol{\theta}^T\mathbf{H}^T\mathbf{H}\boldsymbol{\theta} \\ &= \mathbf{x}^T\mathbf{x} - 2\mathbf{x}^T\mathbf{H}\boldsymbol{\theta} + \boldsymbol{\theta}^T\mathbf{H}^T\mathbf{H}\boldsymbol{\theta} \end{aligned}$$

(note that $\mathbf{x}^T\mathbf{H}\boldsymbol{\theta}$ is a scalar), the gradient is

$$\frac{\partial J(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = -2\mathbf{H}^T\mathbf{x} + 2\mathbf{H}^T\mathbf{H}\boldsymbol{\theta}.$$

Setting the gradient equal to zero yields the LSE

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}. \tag{8.10}$$

The equations $\mathbf{H}^T\mathbf{H}\boldsymbol{\theta} = \mathbf{H}^T\mathbf{x}$ to be solved for $\hat{\boldsymbol{\theta}}$ are termed the *normal equations*. The assumed full rank of $\mathbf{H}$ guarantees the invertibility of $\mathbf{H}^T\mathbf{H}$. See also Problem 8.4 for another derivation. Somewhat surprisingly, we obtain an estimator that has the *identical functional form* as the efficient estimator for the linear model as well as the BLUE. That $\hat{\boldsymbol{\theta}}$ as given by (8.10) is not the *identical estimator* stems from the assumptions made about the data. For it to be the BLUE would require $E(\mathbf{x}) = \mathbf{H}\boldsymbol{\theta}$ and $\mathbf{C}_x = \sigma^2\mathbf{I}$ (see Chapter 6), and to be efficient would in addition to these properties require $\mathbf{x}$ to be Gaussian (see Chapter 4). As a side issue, if these assumptions hold, we can easily determine the statistical properties of the LSE (see Problem 8.6), having been given in Chapters 4 and 6. Otherwise, this may be quite difficult. The minimum LS error is found from (8.9) and (8.10) as

$$\begin{aligned} J_{\min} &= J(\hat{\boldsymbol{\theta}}) \\ &= (\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}})^T(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}) \\ &= \left(\mathbf{x} - \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}\right)^T \left(\mathbf{x} - \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}\right) \\ &= \mathbf{x}^T\left(\mathbf{I} - \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\right)\left(\mathbf{I} - \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\right)\mathbf{x} \\ &= \mathbf{x}^T\left(\mathbf{I} - \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\right)\mathbf{x}. \end{aligned} \tag{8.11}$$

The last step results from the fact that $\mathbf{I} - \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T$ is an idempotent matrix or it has the property $\mathbf{A}^2 = \mathbf{A}$. Other forms for $J_{\min}$ are

$$J_{\min} = \mathbf{x}^T\mathbf{x} - \mathbf{x}^T\mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x} \tag{8.12}$$

$$= \mathbf{x}^T(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}). \tag{8.13}$$

An extension of the linear LS problem is to *weighted* LS. Instead of minimizing (8.9), we include an $N \times N$ positive definite (and by definition therefore symmetric) weighting matrix $\mathbf{W}$, so that

$$J(\boldsymbol{\theta}) = (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T\mathbf{W}(\mathbf{x} - \mathbf{H}\boldsymbol{\theta}). \tag{8.14}$$

If, for instance, $\mathbf{W}$ is diagonal with diagonal elements $[\mathbf{W}]_{ii} = w_i > 0$, then the LS error for Example 8.1 will be

$$J(A) = \sum_{n=0}^{N-1} w_n(x[n] - A)^2.$$

The rationale for introducing weighting factors into the error criterion is to emphasize the contributions of those data samples that are deemed to be more reliable. Again, considering Example 8.1, if $x[n] = A + w[n]$, where $w[n]$ is zero mean uncorrelated noise with variance σ_n^2, then it is reasonable to choose $w_n = 1/\sigma_n^2$. This choice will result in

the estimator (see Problem 8.8)

$$\hat{A} = \frac{\sum_{n=0}^{N-1} \frac{x[n]}{\sigma_n^2}}{\sum_{n=0}^{N-1} \frac{1}{\sigma_n^2}}. \tag{8.15}$$

This familiar estimator is of course the BLUE since the $w[n]$'s are uncorrelated so that $\mathbf{W} = \mathbf{C}^{-1}$ (see Example 6.2).

The general form of the weighted LSE is readily shown to be

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{W}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{W}\mathbf{x} \tag{8.16}$$

and its minimum LS error is

$$J_{\min} = \mathbf{x}^T \left(\mathbf{W} - \mathbf{W}\mathbf{H}(\mathbf{H}^T\mathbf{W}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{W}\right)\mathbf{x} \tag{8.17}$$

(see Problem 8.9).

8.5 Geometrical Interpretations

We now reexamine the linear LS approach from a geometrical perspective. This has the advantage of more clearly revealing the essence of the approach and leads to additional useful properties and insights into the estimator. Recall the general signal model $\mathbf{s} = \mathbf{H}\boldsymbol{\theta}$. If we denote the columns of $\mathbf{H}$ by $\mathbf{h}_i$, we have

$$\begin{aligned} \mathbf{s} &= \begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 & \cdots & \mathbf{h}_p \end{bmatrix} \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \theta_p \end{bmatrix} \\ &= \sum_{i=1}^{p} \theta_i \mathbf{h}_i \end{aligned}$$

so that the signal model is seen to be a linear combination of the "signal" vectors $\{\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_p\}$.

Example 8.4 - Fourier Analysis

Referring to Example 4.2 (with $M = 1$), we suppose the signal model to be

$$s[n] = a\cos 2\pi f_0 n + b\sin 2\pi f_0 n \qquad n = 0, 1, \ldots, N-1$$

where f_0 is a known frequency and $\boldsymbol{\theta} = [a\, b]^T$ is to be estimated. Then, in vector form we have

$$\begin{bmatrix} s[0] \\ s[1] \\ \vdots \\ s[N-1] \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \cos 2\pi f_0 & \sin 2\pi f_0 \\ \vdots & \vdots \\ \cos 2\pi f_0(N-1) & \sin 2\pi f_0(N-1) \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}. \tag{8.18}$$

It is seen that the columns of $\mathbf{H}$ are composed of the samples of the cosinusoidal and sinusoidal sequences. Alternatively, since

$$\begin{aligned} \mathbf{h}_1 &= \begin{bmatrix} 1 & \cos 2\pi f_0 & \dots & \cos 2\pi f_0(N-1) \end{bmatrix}^T \\ \mathbf{h}_2 &= \begin{bmatrix} 0 & \sin 2\pi f_0 & \dots & \sin 2\pi f_0(N-1) \end{bmatrix}^T, \end{aligned}$$

we have

$$\mathbf{s} = a\mathbf{h}_1 + b\mathbf{h}_2.$$

◇

The LS error was defined to be

$$J(\boldsymbol{\theta}) = (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T(\mathbf{x} - \mathbf{H}\boldsymbol{\theta}).$$

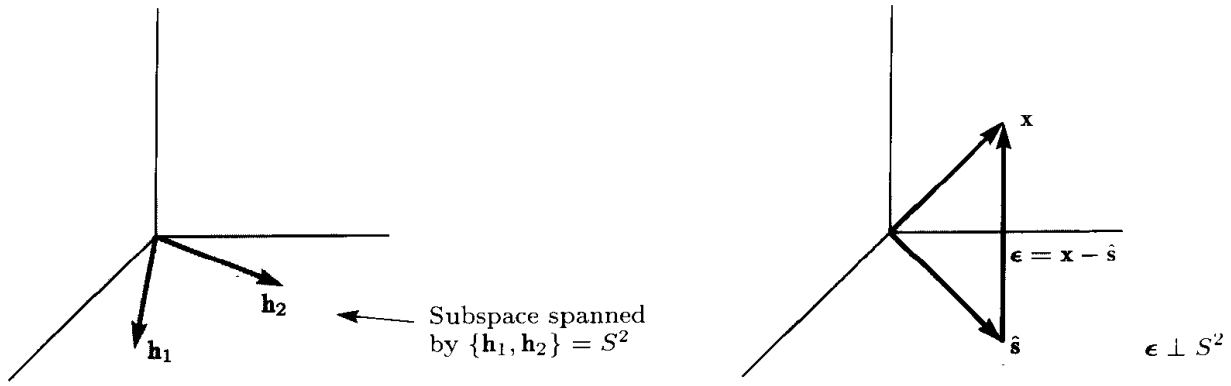
If we further define the Euclidean length of an $N \times 1$ vector $\boldsymbol{\xi} = [\xi_1\, \xi_2 \dots \xi_N]^T$ as

$$||\boldsymbol{\xi}|| = \sqrt{\sum_{i=1}^{N} \xi_i^2} = \sqrt{\boldsymbol{\xi}^T\boldsymbol{\xi}},$$

then the LS error can be also written as

$$\begin{aligned} J(\boldsymbol{\theta}) &= ||\mathbf{x} - \mathbf{H}\boldsymbol{\theta}||^2 \\ &= ||\mathbf{x} - \sum_{i=1}^{p} \theta_i \mathbf{h}_i||^2. \end{aligned} \tag{8.19}$$

We now see that the linear LS approach attempts to minimize the square of the distance from the data vector $\mathbf{x}$ to a signal vector $\sum_{i=1}^{p} \theta_i \mathbf{h}_i$, which must be a linear combination of the columns of $\mathbf{H}$. The data vector can lie anywhere in an N-dimensional space, termed R^N, while all possible signal vectors, being linear combinations of $p < N$ vectors, must lie in a p-dimensional subspace of R^N, termed S^p. (The full rank of $\mathbf{H}$ assumption assures us that the columns are linearly independent and hence the subspace spanned is truly p-dimensional.) For $N = 3$ and $p = 2$ we illustrate this in Figure 8.2. Note that all possible choices of θ_1, θ_2 (where we assume $-\infty < \theta_1 < \infty$ and $-\infty < \theta_2 < \infty$) produce signal vectors constrained to lie in the subspace S^2 and that in general $\mathbf{x}$ does not lie in the subspace. It should be intuitively clear that the vector $\hat{\mathbf{s}}$ that lies in S^2 and



(a) Signal subspace

(b) Orthogonal projection to determine signal estimate

Figure 8.2 Geometrical viewpoint of linear least squares in $\mathbf{R}^3$

that is closest to $\mathbf{x}$ in the Euclidean sense is the component of $\mathbf{x}$ in S^2. Alternatively, $\hat{\mathbf{s}}$ is the *orthogonal projection* of $\mathbf{x}$ onto S^2. This means that the error vector $\mathbf{x} - \hat{\mathbf{s}}$ must be orthogonal to all vectors in S^2. Two vectors in R^N are defined to be orthogonal if $\mathbf{x}^T\mathbf{y} = 0$. To actually determine $\hat{\mathbf{s}}$ for this example we use the orthogonality condition. This says that the error vector is orthogonal to the signal subspace or

$$(\mathbf{x} - \hat{\mathbf{s}}) \perp S^2$$

where $\perp$ denotes orthogonal (or perpendicular). For this to be true we must have

$$\begin{aligned} (\mathbf{x} - \hat{\mathbf{s}}) &\perp \mathbf{h}_1 \\ (\mathbf{x} - \hat{\mathbf{s}}) &\perp \mathbf{h}_2 \end{aligned}$$

since then the error vector will be orthogonal to any linear combination of $\mathbf{h}_1$ and $\mathbf{h}_2$. Using the definition of orthogonality, we have

$$\begin{aligned} (\mathbf{x} - \hat{\mathbf{s}})^T\mathbf{h}_1 &= 0 \\ (\mathbf{x} - \hat{\mathbf{s}})^T\mathbf{h}_2 &= 0. \end{aligned}$$

Letting $\hat{\mathbf{s}} = \theta_1\mathbf{h}_1 + \theta_2\mathbf{h}_2$, we have

$$\begin{aligned} (\mathbf{x} - \theta_1\mathbf{h}_1 - \theta_2\mathbf{h}_2)^T\mathbf{h}_1 &= 0 \\ (\mathbf{x} - \theta_1\mathbf{h}_1 - \theta_2\mathbf{h}_2)^T\mathbf{h}_2 &= 0. \end{aligned}$$

In matrix form this is

$$\begin{aligned} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T\mathbf{h}_1 &= 0 \\ (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T\mathbf{h}_2 &= 0. \end{aligned}$$

Combining the two equations yields

$$(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 \end{bmatrix} = \mathbf{0}^T$$

or

$$(\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{H} = \mathbf{0}^T. \tag{8.20}$$

Finally, we have as our LSE

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}.$$

Note that if $\boldsymbol{\epsilon} = \mathbf{x} - \mathbf{H}\boldsymbol{\theta}$ denotes the error vector, then the LSE is found from (8.20) by invoking the condition

$$\boldsymbol{\epsilon}^T\mathbf{H} = \mathbf{0}^T. \tag{8.21}$$

The error vector must be orthogonal to the columns of $\mathbf{H}$. This is the well-known *orthogonality principle.* In effect, the error represents the part of $\mathbf{x}$ that cannot be described by the signal model. A similar orthogonality principle will arise in Chapter 12 in our study of estimation of random parameters.

Again referring to Figure 8.2b, the minimum LS error is $||\mathbf{x} - \hat{\mathbf{s}}||^2$ or

$$\begin{aligned} ||\mathbf{x} - \hat{\mathbf{s}}||^2 &= ||\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}||^2 \\ &= (\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}})^T(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}). \end{aligned}$$

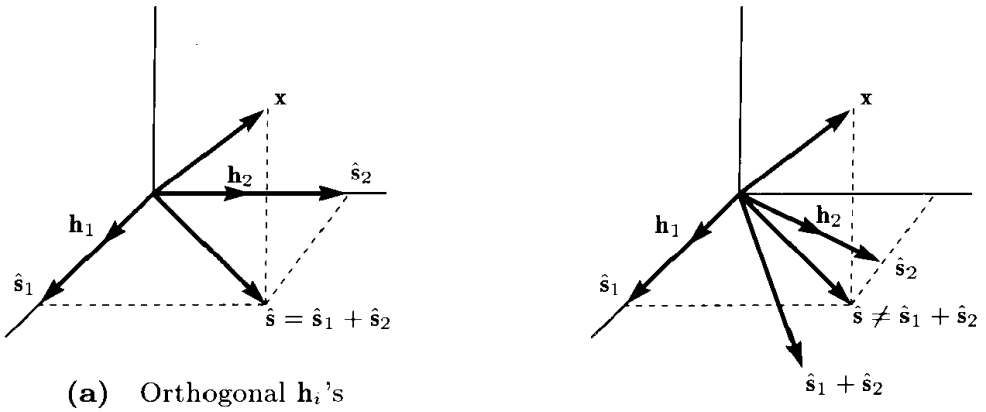
In evaluating this error we can make use of (8.21) (we have already done so for the scalar case in arriving at (8.5)). This produces

$$\begin{aligned} J_{\min} &= (\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}})^T(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}) \\ &= \mathbf{x}^T(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}) - \hat{\boldsymbol{\theta}}^T\mathbf{H}^T\boldsymbol{\epsilon} \\ &= \mathbf{x}^T\mathbf{x} - \mathbf{x}^T\mathbf{H}\hat{\boldsymbol{\theta}} \\ &= \mathbf{x}^T\left(\mathbf{I} - \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\right)\mathbf{x}. \end{aligned} \tag{8.22}$$

In summary, the LS approach can be interpreted as the problem of fitting or approximating a data vector $\mathbf{x}$ in R^N by another vector $\hat{\mathbf{s}}$, which is a linear combination of vectors $\{\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_p\}$ that lie in a p-dimensional subspace of R^N. The problem is solved by choosing $\hat{\mathbf{s}}$ in the subspace to be the orthogonal projection of $\mathbf{x}$. Many of our intuitive notions about vector geometry may be used to our advantage once this connection is made. We now discuss some of these consequences.

Referring to Figure 8.3a, if it had happened that $\mathbf{h}_1$ and $\mathbf{h}_2$ were orthogonal, then $\hat{\mathbf{s}}$ could have easily been found. This is because the component of $\hat{\mathbf{s}}$ along $\mathbf{h}_1$ or $\hat{\mathbf{s}}_1$ does not contain a component of $\hat{\mathbf{s}}$ along $\mathbf{h}_2$. If it did, then we would have the situation in Figure 8.3b. Making the orthogonality assumption and also assuming that $||\mathbf{h}_1|| = ||\mathbf{h}_2|| = 1$ (ortho*normal* vectors), we have

$$\begin{aligned} \hat{\mathbf{s}} &= \hat{\mathbf{s}}_1 + \hat{\mathbf{s}}_2 \\ &= (\mathbf{h}_1^T\mathbf{x})\mathbf{h}_1 + (\mathbf{h}_2^T\mathbf{x})\mathbf{h}_2 \end{aligned}$$



(a) Orthogonal $\mathbf{h}_i$'s

(b) Nonorthogonal $\mathbf{h}_i$'s

Figure 8.3 Effect of nonorthogonal columns of observation matrix

where $\mathbf{h}_i^T\mathbf{x}$ is the length of the vector $\mathbf{x}$ along $\mathbf{h}_i$. In matrix notation this is

$$\begin{aligned}\hat{\mathbf{s}} &= \begin{bmatrix}\mathbf{h}_1 & \mathbf{h}_2\end{bmatrix}\begin{bmatrix}\mathbf{h}_1^T\mathbf{x} \\ \mathbf{h}_2^T\mathbf{x}\end{bmatrix} \\ &= \begin{bmatrix}\mathbf{h}_1 & \mathbf{h}_2\end{bmatrix}\begin{bmatrix}\mathbf{h}_1^T \\ \mathbf{h}_2^T\end{bmatrix}\mathbf{x} \\ &= \mathbf{H}\mathbf{H}^T\mathbf{x}\end{aligned}$$

so that

$$\hat{\boldsymbol{\theta}} = \mathbf{H}^T\mathbf{x}.$$

This result is due to the orthonormal columns of $\mathbf{H}$. As a result, we have

$$(\mathbf{H}^T\mathbf{H})^{-1} = (\mathbf{I})^{-1} = \mathbf{I}$$

and therefore

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x} = \mathbf{H}^T\mathbf{x}.$$

No inversion is necessary. An example follows.

Example 8.5 - Fourier Analysis (continued)

Continuing Example 8.4, if $f_0 = k/N$, where k is an integer taking on any of the values $k = 1, 2, \ldots, N/2 - 1$, it is easily shown (see (4.13)) that

$$\mathbf{h}_1^T\mathbf{h}_2 = \sum_{n=0}^{N-1} \cos\left(2\pi\frac{k}{N}n\right)\sin\left(2\pi\frac{k}{N}n\right) = 0$$

and also

$$\begin{aligned} \mathbf{h}_1^T\mathbf{h}_1 &= \frac{N}{2} \\ \mathbf{h}_2^T\mathbf{h}_2 &= \frac{N}{2} \end{aligned}$$

so that $\mathbf{h}_1$ and $\mathbf{h}_2$ are orthogonal but not orthonormal. Combining these results produces $\mathbf{H}^T\mathbf{H} = (N/2)\mathbf{I}$, and therefore,

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \begin{bmatrix} \hat{a} \\ \hat{b} \end{bmatrix} \\ &= (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x} \\ &= \frac{2}{N}\mathbf{H}^T\mathbf{x} \\ &= \begin{bmatrix} \frac{2}{N}\sum_{n=0}^{N-1} x[n]\cos\left(2\pi\frac{k}{N}n\right) \\ \frac{2}{N}\sum_{n=0}^{N-1} x[n]\sin\left(2\pi\frac{k}{N}n\right) \end{bmatrix}. \end{aligned}$$

If we had instead defined the signal as

$$s[n] = a'\sqrt{\frac{2}{N}}\cos\left(2\pi\frac{k}{N}n\right) + b'\sqrt{\frac{2}{N}}\sin\left(2\pi\frac{k}{N}n\right)$$

then the columns of $\mathbf{H}$ would have been orthonormal. ◇

In general, the columns of $\mathbf{H}$ will not be orthogonal, so that the signal vector estimate is obtained as

$$\hat{\mathbf{s}} = \mathbf{H}\hat{\boldsymbol{\theta}} = \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}.$$

The signal estimate is the orthogonal projection of $\mathbf{x}$ onto the p-dimensional subspace. The $N \times N$ matrix $\mathbf{P} = \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T$ is known as the *orthogonal projection matrix* or just the *projection matrix*. It has the properties

1. $\mathbf{P}^T = \mathbf{P}$, symmetric
2. $\mathbf{P}^2 = \mathbf{P}$, idempotent.

That the projection matrix must be symmetric is shown in Problem 8.11, that it must be idempotent follows from the observation that if $\mathbf{P}$ is applied to $\mathbf{Px}$, then the same vector must result since $\mathbf{Px}$ is already in the subspace. Additionally, the projection matrix must be singular (for independent columns of $\mathbf{H}$ it has rank p, as shown in Problem 8.12). If it were not, then $\mathbf{x}$ could be recovered from $\hat{\mathbf{s}}$, which is clearly impossible since many $\mathbf{x}$'s have the same projection, as shown in Figure 8.4.

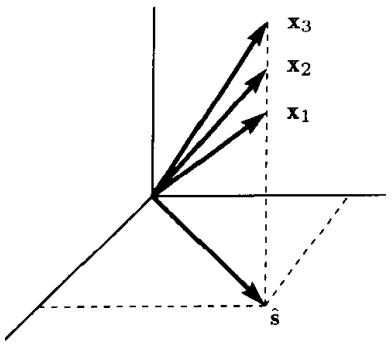


Figure 8.4 Vectors with same projection onto subspace

Likewise, the error vector $\epsilon = \mathbf{x} - \hat{\mathbf{s}} = (\mathbf{I} - \mathbf{P})\mathbf{x}$ is the projection of $\mathbf{x}$ onto the *complement* subspace or the subspace orthogonal to the signal subspace. The matrix $\mathbf{P}^{\perp} = \mathbf{I} - \mathbf{P}$ is also a projection matrix, as can be easily verified from the properties given above. As a result, the minimum LS error is, from (8.22),

$$\begin{aligned} J_{\min} &= \mathbf{x}^T(\mathbf{I} - \mathbf{P})\mathbf{x} \\ &= \mathbf{x}^T\mathbf{P}^{\perp}\mathbf{x} \\ &= \mathbf{x}^T\mathbf{P}^{\perp T}\mathbf{P}^{\perp}\mathbf{x} \\ &= ||\mathbf{P}^{\perp}\mathbf{x}||^2 \end{aligned}$$

which is just $||\epsilon||^2$.

In the next section we further utilize the geometrical theory to derive an order-recursive LS solution.

8.6 Order-Recursive Least Squares

In many cases the signal model is unknown and must be assumed. For example, consider the experimental data shown in Figure 8.5a. The following models might be assumed

$$\begin{aligned} s_1(t) &= A \\ s_2(t) &= A + Bt \end{aligned}$$

for $0 \leq t \leq T$. If data $x[n]$ are obtained by sampling $x(t)$ at times $t = n\Delta$, where $\Delta = 1$ and $n = 0, 1, \ldots, N-1$, the corresponding discrete-time signal models become

$$\begin{aligned} s_1[n] &= A \\ s_2[n] &= A + Bn. \end{aligned}$$

Using a LSE with

$$\mathbf{H}_1 = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \qquad \mathbf{H}_2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & N-1 \end{bmatrix}$$

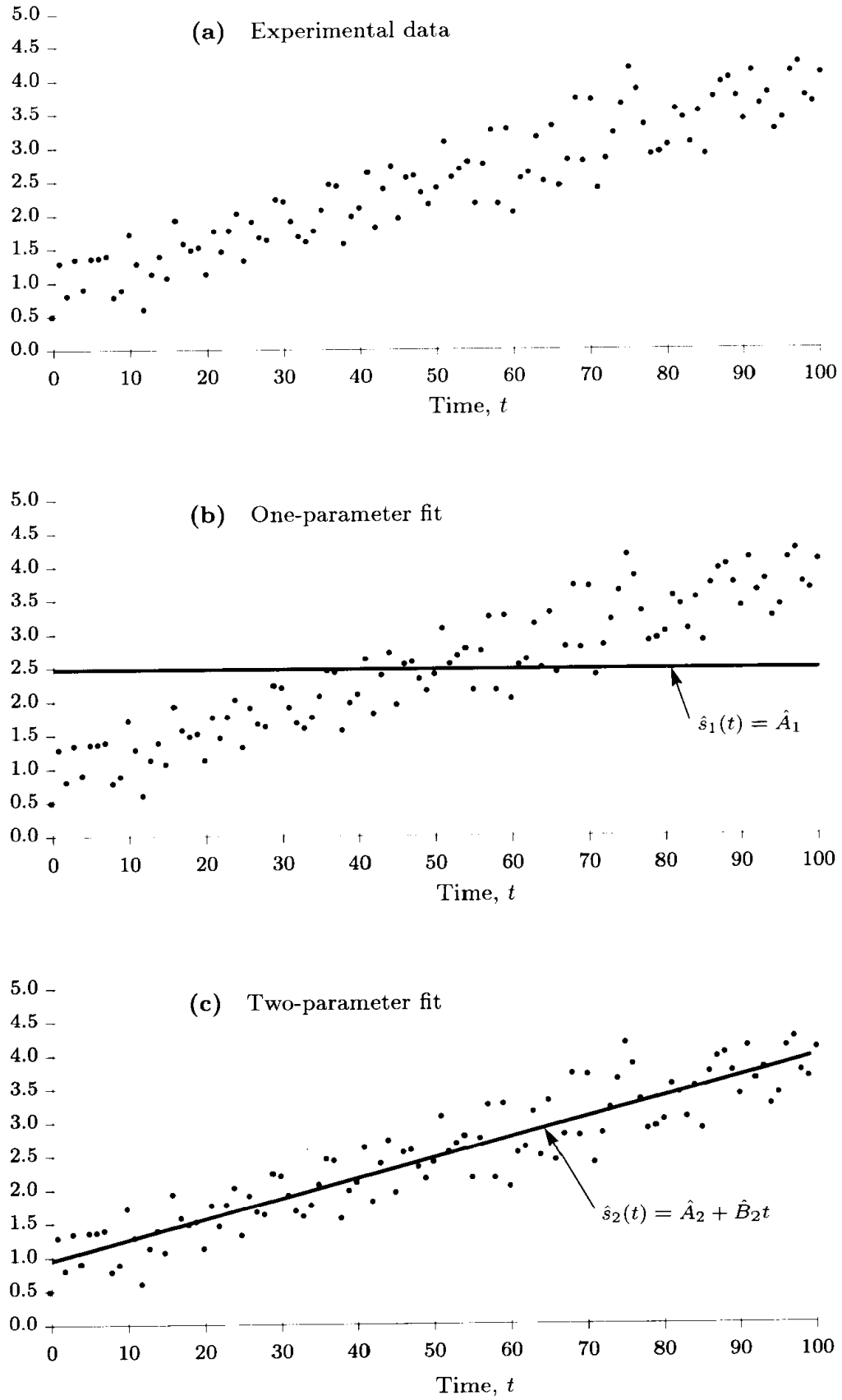


Figure 8.5 Experimental data fitting by least squares

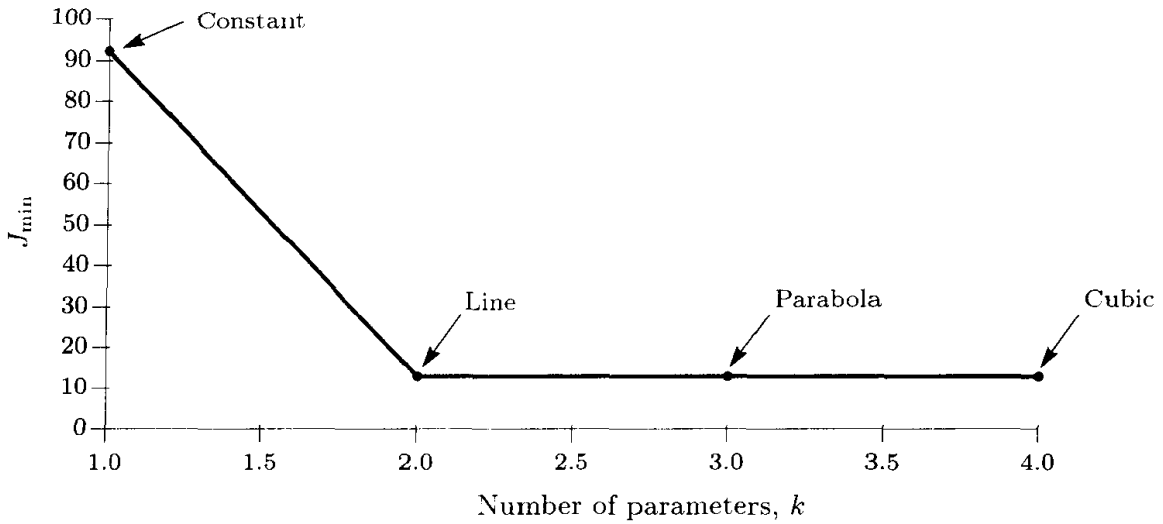


Figure 8.6 Effect of chosen number of parameters on minimum least squares error

would produce the estimates for the intercept and slope as

$$\hat{A}_1 = \bar{x} \tag{8.23}$$

and (see Problem 8.13)

$$\begin{aligned} \hat{A}_2 &= \frac{2(2N-1)}{N(N+1)} \sum_{n=0}^{N-1} x[n] - \frac{6}{N(N+1)} \sum_{n=0}^{N-1} nx[n] \\ \hat{B}_2 &= -\frac{6}{N(N+1)} \sum_{n=0}^{N-1} x[n] + \frac{12}{N(N^2-1)} \sum_{n=0}^{N-1} nx[n]. \end{aligned} \tag{8.24}$$

In Figures 8.5b and 8.5c we have plotted $\hat{s}_1(t) = \hat{A}_1$ and $\hat{s}_2(t) = \hat{A}_2 + \hat{B}_2 t$, where $T = 100$. The fit using two parameters is better, as expected. We will later show that the minimum LS error must decrease as we add more parameters. The question might be asked whether or not we should add a quadratic term to the signal model. If we did, the fit would become better yet. Realizing that the data are subject to error, we may very well be *fitting the noise*. Such a situation is undesirable but in the absence of a known signal model may be unavoidable to some extent. In practice, we choose the simplest signal model that adequately describes the data. One such scheme might be to increase the order of the polynomial until the minimum LS error decreases only slightly as the order is increased. For the data in Figure 8.5 the minimum LS error versus the number of parameters is shown in Figure 8.6. Models $s_1(t)$ and $s_2(t)$ correspond to $k = 1$ and $k = 2$, respectively. It is seen that a large drop occurs at $k = 2$. For larger orders there is only a slight decrease, indicating a modeling of the noise. In this example the signal was actually

$$s(t) = 1 + 0.03t$$

and the noise $w[n]$ was WGN with variance $\sigma^2 = 0.1$. Note that if A, B had been estimated perfectly, the minimum LS error would have been

$$\begin{aligned} J_{\min} &= J(\hat{A}, \hat{B}) \\ &= J(A, B) \\ &= \sum_{n=0}^{N-1} (x[n] - s[n])^2 \\ &= \sum_{n=0}^{N-1} w^2[n] \\ &\approx N\sigma^2 \end{aligned}$$

so that when the true order is reached, $J_{\min} \approx 10$. This is verified in Figure 8.6 and also increases our confidence in the chosen model.

In the preceding example we saw that it was important to be able to determine the LSE for several signal models. A straightforward approach would compute the LSE for each model using (8.10). Alternatively, the computation may be reduced by using an *order-recursive* LS approach. In this method we *update* the LSE in order. Specifically, we are able to compute the LSE based on an $\mathbf{H}$ of dimension $N \times (k+1)$ from the solution based on an $\mathbf{H}$ of dimension $N \times k$. For the previous example this update in order would have been exceedingly simple had the columns of $\mathbf{H}_2$ been orthogonal. To see why assume the signal models to be

$$\begin{aligned} s_1[n] &= A_1 \\ s_2[n] &= A_2 + B_2 n \end{aligned}$$

for $-M \leq n \leq M$. We have now altered the observation interval to be the symmetric interval $[-M, M]$ as opposed to the original $[0, N-1]$ interval. The effect of this assumption is to orthogonalize the columns of $\mathbf{H}_2$ since now

$$\mathbf{H}_2 = \begin{bmatrix} 1 & -M \\ 1 & -(M-1) \\ \vdots & \vdots \\ 1 & M \end{bmatrix}.$$

The LSE is readily found since

$$\mathbf{H}_2^T \mathbf{H}_2 = \begin{bmatrix} 2M+1 & 0 \\ 0 & \sum_{n=-M}^{M} n^2 \end{bmatrix}$$

is a diagonal matrix. The solutions are

$$\hat{A}_1 = \frac{1}{2M+1} \sum_{n=-M}^{M} x[n]$$

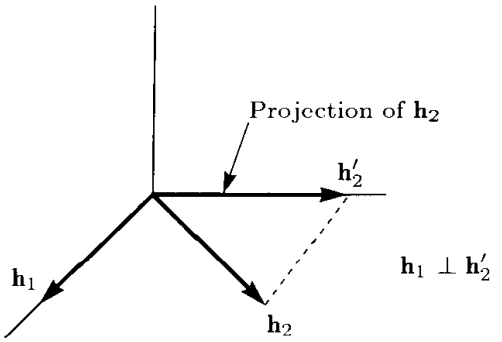


Figure 8.7 Gram-Schmidt orthogonalization

and

$$\hat{A}_2 = \frac{1}{2M+1} \sum_{n=-M}^{M} x[n]$$

$$\hat{B}_2 = \frac{\sum_{n=-M}^{M} nx[n]}{\sum_{n=-M}^{M} n^2}.$$

In this case the LSE of A does not change as we add a parameter to the model and follows algebraically from the diagonal nature of $\mathbf{H}_2^T\mathbf{H}_2$. In geometric terms this result is readily apparent from Figure 8.3a. The orthogonality of the $\mathbf{h}_i$'s allows us to project $\mathbf{x}$ along $\mathbf{h}_1$ and $\mathbf{h}_2$ separately and then add the results. Each projection is independent of the other. In general, the column vectors will not be orthogonal but can be replaced by another set of p vectors that are orthogonal. The procedure for doing so is shown in Figure 8.7 and is called a *Gram-Schmidt orthogonalization.* The new column $\mathbf{h}_2$ is projected onto the subspace *orthogonal* to $\mathbf{h}_1$. Then, since $\mathbf{h}_1$ and $\mathbf{h}_2'$ are orthogonal, the LS signal estimate becomes

$$\hat{\mathbf{s}} = \mathbf{h}_1\hat{\theta}_1 + \mathbf{h}_2'\hat{\theta}_2'$$

where $\mathbf{h}_1\hat{\theta}_1$ is also the LSE for the signal based on $\mathbf{H} = \mathbf{h}_1$ only. This procedure recursively adds terms to the signal model; it updates the order. In Problem 8.14 this geometrical viewpoint is used together with a Gram-Schmidt orthogonalization to derive the order-update equations. A purely algebraic derivation is given in Appendix 8A. It is now summarized.

Denote the $N \times k$ observation matrix as $\mathbf{H}_k$, and the LSE based on $\mathbf{H}_k$ as $\hat{\boldsymbol{\theta}}_k$ or

$$\hat{\boldsymbol{\theta}}_k = (\mathbf{H}_k^T\mathbf{H}_k)^{-1}\mathbf{H}_k^T\mathbf{x}. \tag{8.25}$$

The minimum LS error based on $\mathbf{H}_k$ is

$$J_{\min_k} = (\mathbf{x} - \mathbf{H}_k\hat{\boldsymbol{\theta}}_k)^T(\mathbf{x} - \mathbf{H}_k\hat{\boldsymbol{\theta}}_k). \tag{8.26}$$

By increasing the order to $k+1$ we add a column to the observation matrix. This generates a new observation matrix which in partitioned form is

$$\mathbf{H}_{k+1} = \left[\begin{array}{cc} \mathbf{H}_k & \mathbf{h}_{k+1} \end{array} \right] = \left[\begin{array}{cc} N \times k & N \times 1 \end{array} \right]. \tag{8.27}$$

To update $\hat{\boldsymbol{\theta}}_k$ and $J_{\min_k}$ we use

$$\hat{\boldsymbol{\theta}}_{k+1} = \left[\begin{array}{c} \hat{\boldsymbol{\theta}}_k - \dfrac{(\mathbf{H}_k^T\mathbf{H}_k)^{-1}\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}} \\ \dfrac{\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}} \end{array} \right] = \left[\begin{array}{c} k \times 1 \\ 1 \times 1 \end{array} \right] \tag{8.28}$$

where

$$\mathbf{P}_k^{\perp} = \mathbf{I} - \mathbf{H}_k(\mathbf{H}_k^T\mathbf{H}_k)^{-1}\mathbf{H}_k^T$$

is the projection matrix onto the subspace orthogonal to that spanned by the columns of $\mathbf{H}_k$. To avoid inverting $\mathbf{H}_k^T\mathbf{H}_k$ we let

$$\mathbf{D}_k = (\mathbf{H}_k^T\mathbf{H}_k)^{-1} \tag{8.29}$$

and use the recursive formula

$$\begin{aligned} \mathbf{D}_{k+1} &= \left[\begin{array}{cc} \mathbf{D}_k + \dfrac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k}{\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}} & -\dfrac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}} \\ -\dfrac{\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k}{\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}} & \dfrac{1}{\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}} \end{array} \right] \\ &= \left[\begin{array}{cc} k \times k & k \times 1 \\ 1 \times k & 1 \times 1 \end{array} \right] \end{aligned} \tag{8.30}$$

where $\mathbf{P}_k^{\perp} = \mathbf{I} - \mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T$. The minimum LS error is updated by using

$$J_{\min_{k+1}} = J_{\min_k} - \frac{\left(\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{x}\right)^2}{\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}}. \tag{8.31}$$

The entire algorithm requires no matrix inversions. The recursion begins by determining $\hat{\theta}_1$, $J_{\min_1}$, and $\mathbf{D}_1$ using (8.25), (8.26), and (8.29), respectively. We term (8.28), (8.30), and (8.31) the *order-recursive least squares method.* To illustrate the computations involved we apply the method to the previous line fitting example.

Example 8.6 - Line Fitting

Since $s_1[n] = A_1$ and $s_2[n] = A_2 + B_2 n$ for $n = 0, 1, \ldots, N-1$, we have

$$\begin{aligned} \mathbf{H}_1 &= \left[\begin{array}{c} 1 \\ 1 \\ \vdots \\ 1 \end{array} \right] = \mathbf{1} \\ \mathbf{H}_2 &= \left[\begin{array}{cc} \mathbf{H}_1 & \mathbf{h}_2 \end{array} \right] \end{aligned}$$

where

$$\mathbf{h}_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ N-1 \end{bmatrix}.$$

Using (8.25) and (8.26) to begin the recursion

$$\begin{aligned} \hat{A}_1 &= \hat{\theta}_1 \\ &= (\mathbf{H}_1^T\mathbf{H}_1)^{-1}\mathbf{H}_1^T\mathbf{x} \\ &= \bar{x} \end{aligned}$$

and

$$\begin{aligned} J_{\min_1} &= (\mathbf{x} - \mathbf{H}_1\hat{\theta}_1)^T(\mathbf{x} - \mathbf{H}_1\hat{\theta}_1) \\ &= \sum_{n=0}^{N-1}(x[n] - \bar{x})^2. \end{aligned}$$

Next, we use (8.28) to find $\hat{\boldsymbol{\theta}}_2 = [\hat{A}_2\ \hat{B}_2]^T$ or

$$\hat{\boldsymbol{\theta}}_2 = \begin{bmatrix} \hat{\theta}_1 - \dfrac{(\mathbf{H}_1^T\mathbf{H}_1)^{-1}\mathbf{H}_1^T\mathbf{h}_2\mathbf{h}_2^T\mathbf{P}_1^\perp\mathbf{x}}{\mathbf{h}_2^T\mathbf{P}_1^\perp\mathbf{h}_2} \\ \dfrac{\mathbf{h}_2^T\mathbf{P}_1^\perp\mathbf{x}}{\mathbf{h}_2^T\mathbf{P}_1^\perp\mathbf{h}_2} \end{bmatrix}.$$

The necessary terms are

$$\begin{aligned} (\mathbf{H}_1^T\mathbf{H}_1)^{-1} &= \frac{1}{N} \\ \mathbf{P}_1^\perp &= \mathbf{I} - \mathbf{H}_1(\mathbf{H}_1^T\mathbf{H}_1)^{-1}\mathbf{H}_1^T \\ &= \mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^T \\ \mathbf{P}_1^\perp\mathbf{x} &= \mathbf{x} - \frac{1}{N}\mathbf{1}\mathbf{1}^T\mathbf{x} \\ &= \mathbf{x} - \bar{x}\mathbf{1} \\ \mathbf{h}_2^T\mathbf{P}_1^\perp\mathbf{x} &= \mathbf{h}_2^T\mathbf{x} - \bar{x}\mathbf{h}_2^T\mathbf{1} \\ &= \sum_{n=0}^{N-1} nx[n] - \bar{x}\sum_{n=0}^{N-1} n \\ \mathbf{h}_2^T\mathbf{P}_1^\perp\mathbf{h}_2 &= \mathbf{h}_2^T\mathbf{h}_2 - \frac{1}{N}(\mathbf{h}_2^T\mathbf{1})^2 \\ &= \sum_{n=0}^{N-1} n^2 - \frac{1}{N}\left(\sum_{n=0}^{N-1} n\right)^2. \end{aligned}$$

Upon substitution we have

$$\hat{\boldsymbol{\theta}}_2 = \begin{bmatrix} \bar{x} - \dfrac{\dfrac{1}{N}\displaystyle\sum_{n=0}^{N-1} n\left[\sum_{n=0}^{N-1} nx[n] - \bar{x}\sum_{n=0}^{N-1} n\right]}{\displaystyle\sum_{n=0}^{N-1} n^2 - \frac{1}{N}\left(\sum_{n=0}^{N-1} n\right)^2} \\ \dfrac{\displaystyle\sum_{n=0}^{N-1} nx[n] - \bar{x}\sum_{n=0}^{N-1} n}{\displaystyle\sum_{n=0}^{N-1} n^2 - \frac{1}{N}\left(\sum_{n=0}^{N-1} n\right)^2} \end{bmatrix}$$

$$= \begin{bmatrix} \bar{x} - \dfrac{1}{N}\displaystyle\sum_{n=0}^{N-1} n\hat{B}_2 \\ \hat{B}_2 \end{bmatrix}.$$

Simplifying produces

$$\hat{B}_2 = \frac{\displaystyle\sum_{n=0}^{N-1} nx[n] - \bar{x}\sum_{n=0}^{N-1} n}{\displaystyle\sum_{n=0}^{N-1} n^2 - \frac{1}{N}\left(\sum_{n=0}^{N-1} n\right)^2}$$

$$= \frac{\displaystyle\sum_{n=0}^{N-1} nx[n] - \frac{N(N-1)}{2}\bar{x}}{N(N^2-1)/12}$$

$$= -\frac{6}{N(N+1)}\sum_{n=0}^{N-1} x[n] + \frac{12}{N(N^2-1)}\sum_{n=0}^{N-1} nx[n]$$

and

$$\hat{A}_2 = \bar{x} - \frac{1}{N}\sum_{n=0}^{N-1} n\hat{B}_2$$

$$= \bar{x} - \frac{N-1}{2}\hat{B}_2$$

$$= \frac{2(2N-1)}{N(N+1)}\sum_{n=0}^{N-1} x[n] - \frac{6}{N(N+1)}\sum_{n=0}^{N-1} nx[n].$$

These are seen to agree with (8.24). The minimum LS error is, from (8.31),

$$\begin{aligned} J_{\min_2} &= J_{\min_1} - \frac{(\mathbf{h}_2^T\mathbf{P}_1^{\perp}\mathbf{x})^2}{\mathbf{h}_2^T\mathbf{P}_1^{\perp}\mathbf{h}_2} \\ &= J_{\min_1} - \frac{\left(\sum_{n=0}^{N-1} nx[n] - \bar{x}\sum_{n=0}^{N-1} n\right)^2}{\sum_{n=0}^{N-1} n^2 - \frac{1}{N}\left(\sum_{n=0}^{N-1} n\right)^2} \\ &= J_{\min_1} - \frac{\left(\sum_{n=0}^{N-1} nx[n] - \frac{N}{2}(N-1)\bar{x}\right)^2}{N(N^2-1)/12}. \end{aligned}$$

◇

In solving for the LSE of the parameters of the second-order model we first solved for the LSE of the first-order model. In general, the recursive procedure solves successive LS problems until the desired order is attained. Hence, the recursive order-update solution not only determines the LSE for the desired model but also *determines the LSE for all lower-order models as well.* As discussed previously, this property is useful when the model order is not known a priori.

Several interesting observations can be made based on (8.28).

1. If the new column $\mathbf{h}_{k+1}$ is orthogonal to all the previous ones, then $\mathbf{H}_k^T\mathbf{h}_{k+1} = \mathbf{0}$ and from (8.28)

$$\hat{\boldsymbol{\theta}}_{k+1} = \begin{bmatrix} \hat{\boldsymbol{\theta}}_k \\ \dfrac{\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}} \end{bmatrix}$$

   or the LSE for the first k components of $\hat{\boldsymbol{\theta}}_{k+1}$ remains the same. This generalizes the geometrical illustration in Figure 8.3a.

2. The term $\mathbf{P}_k^{\perp}\mathbf{x} = (\mathbf{I} - \mathbf{H}_k(\mathbf{H}_k^T\mathbf{H}_k)^{-1}\mathbf{H}_k^T)\mathbf{x} = \mathbf{x} - \mathbf{H}_k\hat{\boldsymbol{\theta}}_k = \boldsymbol{\epsilon}_k$ is the LS error vector or the *data residual* that cannot be modeled by the columns of $\mathbf{H}_k$. It represents the component of $\mathbf{x}$ orthogonal to the space spanned by the columns of $\mathbf{H}_k$ (see Problem 8.17). We can view $\mathbf{P}_k^{\perp}\mathbf{x}$, the residual, as the part of $\mathbf{x}$ yet to be modeled.

3. If $\mathbf{h}_{k+1}$ is nearly in the space spanned by the columns of $\mathbf{H}_k$, then, as shown in Figure 8.8, $\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}$ will be small. Consequently,

$$\begin{aligned} \mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1} &= \mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp^T}\mathbf{P}_k^{\perp}\mathbf{h}_{k+1} \\ &= ||\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}||^2 \\ &\approx 0 \end{aligned}$$

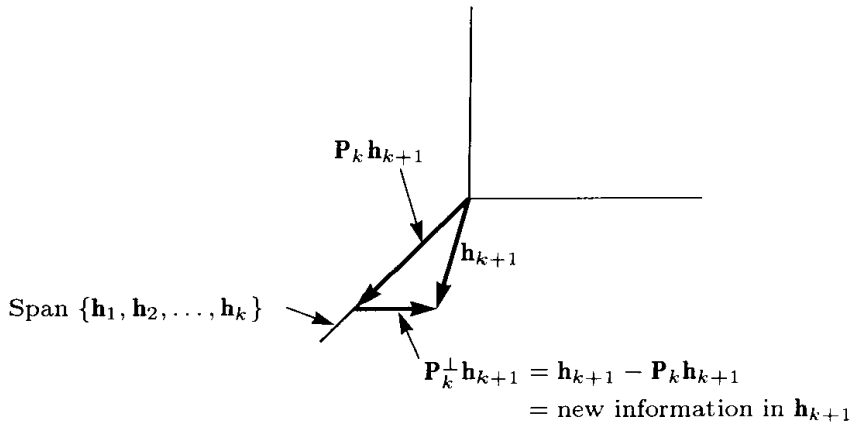


Figure 8.8 Near collinearity of columns of observation matrix

and the recursive procedure will "blow up" (see (8.28)). In essence, since $\mathbf{h}_{k+1}$ nearly lies in the subspace spanned by the columns of $\mathbf{H}_k$, the new observation matrix $\mathbf{H}_{k+1}$ will be nearly of rank k and hence $\mathbf{H}_{k+1}^T\mathbf{H}_{k+1}$ will be nearly singular. In practice, we could monitor the term $\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}$ and exclude from the recursion those column vectors that produce small values of this term.

4. The minimum LS error can also be written in a more suggestive fashion, indicating the contribution of the new parameter in reducing the error. From (8.22) and (8.31) we have

$$\begin{aligned} J_{\min_{k+1}} &= \mathbf{x}^T\mathbf{P}_k^{\perp}\mathbf{x} - \frac{\left(\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{x}\right)^2}{\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}} \\ &= \mathbf{x}^T\mathbf{P}_k^{\perp}\mathbf{x}\left[1 - \frac{\left(\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{x}\right)^2}{\mathbf{x}^T\mathbf{P}_k^{\perp}\mathbf{x}\,\mathbf{h}_{k+1}^T\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}}\right] \\ &= J_{\min_k}(1 - r_{k+1}^2) \end{aligned} \tag{8.32}$$

where

$$r_{k+1}^2 = \frac{\left[(\mathbf{P}_k^{\perp}\mathbf{h}_{k+1})^T(\mathbf{P}_k^{\perp}\mathbf{x})\right]^2}{||\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}||^2||\mathbf{P}_k^{\perp}\mathbf{x}||^2}. \tag{8.33}$$

Letting $(\mathbf{x}, \mathbf{y}) = \mathbf{x}^T\mathbf{y}$ denote the inner product in R^N, we have

$$r_{k+1}^2 = \frac{\left(\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}, \mathbf{P}_k^{\perp}\mathbf{x}\right)^2}{||\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}||^2||\mathbf{P}_k^{\perp}\mathbf{x}||^2}$$

where r_{k+1}^2 is seen to be the square of a correlation coefficient and as such has the property

$$0 \leq r_{k+1}^2 \leq 1.$$

Intuitively, $\mathbf{P}_k^{\perp}\mathbf{x}$ represents the residual or error in modeling $\mathbf{x}$ based on k parameters, while $\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}$ represents the new model information contributed by the $(k+1)$st parameter (see Figure 8.8). For instance, if $\mathbf{P}_k^{\perp}\mathbf{x}$ and $\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}$ are collinear, then $r_{k+1}^2 = 1$ and $J_{\min_{k+1}} = 0$. This says that the part of $\mathbf{x}$ that could not be modeled by the columns of $\mathbf{H}_k$ can be perfectly modeled by $\mathbf{h}_{k+1}$. Assuming that $r_{k+1} \neq 0$, the expression of (8.32) also shows that the minimum LS error monotonically decreases with order as in Figure 8.6.

5. Recall that the LS signal estimate is

$$\hat{\mathbf{s}} = \mathbf{H}\hat{\boldsymbol{\theta}} = \mathbf{P}\mathbf{x}$$

where $\mathbf{P}$ is the projection matrix. It is shown in Appendix 8B that $\mathbf{P}$ may be recursively updated by using (8.28). The result is

$$\mathbf{P}_{k+1} = \mathbf{P}_k + \frac{(\mathbf{I}-\mathbf{P}_k)\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T(\mathbf{I}-\mathbf{P}_k)}{\mathbf{h}_{k+1}^T(\mathbf{I}-\mathbf{P}_k)\mathbf{h}_{k+1}}. \tag{8.34}$$

This is termed the *recursive orthogonal projection matrix*. See Problem 8.18 for its utility in determining the recursive formula for the minimum LS error. If we define the unit length vector

$$\begin{aligned}\mathbf{u}_{k+1} &= \frac{(\mathbf{I}-\mathbf{P}_k)\mathbf{h}_{k+1}}{||(\mathbf{I}-\mathbf{P}_k)\mathbf{h}_{k+1}||} \\ &= \frac{\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}}{||\mathbf{P}_k^{\perp}\mathbf{h}_{k+1}||}\end{aligned}$$

the recursive projection operator becomes

$$\mathbf{P}_{k+1} = \mathbf{P}_k + \mathbf{u}_{k+1}\mathbf{u}_{k+1}^T$$

where $\mathbf{u}_{k+1}$ points in the direction of the new information (see Figure 8.8), or by letting $\hat{\mathbf{s}}_k = \mathbf{P}_k\mathbf{x}$, we have

$$\begin{aligned}\hat{\mathbf{s}}_{k+1} &= \mathbf{P}_{k+1}\mathbf{x} \\ &= \mathbf{P}_k\mathbf{x} + (\mathbf{u}_{k+1}^T\mathbf{x})\mathbf{u}_{k+1} \\ &= \hat{\mathbf{s}}_k + (\mathbf{u}_{k+1}^T\mathbf{x})\mathbf{u}_{k+1}.\end{aligned}$$

8.7 Sequential Least Squares

In many signal processing applications the received data are obtained by sampling a continuous-time waveform. Such data are on-going in that as time progresses, more data become available. We have the option of either waiting for all the available data or, as we now describe, of processing the data sequentially in time. This yields a sequence of LSEs in time. Specifically, assume we have determined the LSE $\hat{\boldsymbol{\theta}}$ based on

$\{x[0], x[1], \ldots, x[N-1]\}$. If we now observe $x[N]$, can we update $\hat{\boldsymbol{\theta}}$ (in time) without having to resolve the linear equations of (8.10)? The answer is yes, and the procedure is termed *sequential least squares* to distinguish it from our original approach in which we processed all the data at once, termed the *batch* approach.

Consider Example 8.1 in which the DC signal level is to be estimated. We saw that the LSE is

$$\hat{A}[N-1] = \frac{1}{N} \sum_{n=0}^{N-1} x[n]$$

where the argument of $\hat{A}$ denotes the index of the most recent data point observed. If we now observe the new data sample $x[N]$, then the LSE becomes

$$\hat{A}[N] = \frac{1}{N+1} \sum_{n=0}^{N} x[n].$$

In computing this new estimator we do not have to recompute the sum of the observations since

$$\begin{aligned} \hat{A}[N] &= \frac{1}{N+1} \left(\sum_{n=0}^{N-1} x[n] + x[N] \right) \\ &= \frac{N}{N+1} \hat{A}[N-1] + \frac{1}{N+1} x[N]. \end{aligned} \tag{8.35}$$

The new LSE is found by using the previous one and the new observation. The sequential approach also lends itself to an interesting interpretation. Rearranging (8.35), we have

$$\hat{A}[N] = \hat{A}[N-1] + \frac{1}{N+1} \left(x[N] - \hat{A}[N-1] \right). \tag{8.36}$$

The new estimate is equal to the old one plus a *correction* term. The correction term decreases with N, reflecting the fact that the estimate $\hat{A}[N-1]$ is based on many more data samples and therefore should be given more weight. Also, $x[N] - \hat{A}[N-1]$ can be thought of as the error in predicting $x[N]$ by the previous samples, which are summarized by $\hat{A}[N-1]$. If this error is zero, then no correction takes place for that update. Otherwise, the new estimate differs from the old one.

The minimum LS error may also be computed recursively. Based on data samples up to time $N-1$, the error is

$$J_{\min}[N-1] = \sum_{n=0}^{N-1} (x[n] - \hat{A}[N-1])^2$$

and thus using (8.36)

$$J_{\min}[N] = \sum_{n=0}^{N} (x[n] - \hat{A}[N])^2$$

$$= \sum_{n=0}^{N-1} \left[x[n] - \hat{A}[N-1] - \frac{1}{N+1}(x[N] - \hat{A}[N-1]) \right]^2 + (x[N] - \hat{A}[N])^2$$

$$= J_{\min}[N-1] - \frac{2}{N+1} \sum_{n=0}^{N-1} (x[n] - \hat{A}[N-1])(x[N] - \hat{A}[N-1])$$

$$+ \frac{N}{(N+1)^2}(x[N] - \hat{A}[N-1])^2 + (x[N] - \hat{A}[N])^2.$$

Noting that the middle term on the right-hand side is zero, we have after some simplication

$$J_{\min}[N] = J_{\min}[N-1] + \frac{N}{N+1}(x[N] - \hat{A}[N-1])^2.$$

The apparent paradoxical behavior of an increase in the minimum LS error is readily explained if we note that with each new data point, the number of squared error terms increases. Thus, we need to fit more points with the same number of parameters.

A more interesting example of a sequential LS approach arises in the *weighted* LS problem. For the present example, if the weighting matrix $\mathbf{W}$ is diagonal, with $[\mathbf{W}]_{ii} = 1/\sigma_i^2$, then the weighted LSE is, from (8.15),

$$\hat{A}[N-1] = \frac{\sum_{n=0}^{N-1} \frac{x[n]}{\sigma_n^2}}{\sum_{n=0}^{N-1} \frac{1}{\sigma_n^2}}.$$

To find the sequential version

$$\hat{A}[N] = \frac{\sum_{n=0}^{N} \frac{x[n]}{\sigma_n^2}}{\sum_{n=0}^{N} \frac{1}{\sigma_n^2}}$$

$$= \frac{\sum_{n=0}^{N-1} \frac{x[n]}{\sigma_n^2} + \frac{x[N]}{\sigma_N^2}}{\sum_{n=0}^{N} \frac{1}{\sigma_n^2}}$$

$$= \frac{\left(\sum_{n=0}^{N-1} \frac{1}{\sigma_n^2} \right) \hat{A}[N-1]}{\sum_{n=0}^{N} \frac{1}{\sigma_n^2}} + \frac{\frac{x[N]}{\sigma_N^2}}{\sum_{n=0}^{N} \frac{1}{\sigma_n^2}}$$

$$= \hat{A}[N-1] - \frac{\frac{1}{\sigma_N^2}\hat{A}[N-1]}{\sum_{n=0}^{N}\frac{1}{\sigma_n^2}} + \frac{\frac{x[N]}{\sigma_N^2}}{\sum_{n=0}^{N}\frac{1}{\sigma_n^2}}$$

or finally

$$\hat{A}[N] = \hat{A}[N-1] + \frac{\frac{1}{\sigma_N^2}}{\sum_{n=0}^{N}\frac{1}{\sigma_n^2}}(x[N] - \hat{A}[N-1]). \tag{8.37}$$

As expected, if $\sigma_n^2 = \sigma^2$ for all n, we have our previous result. The gain factor that multiplies the correction term now depends on our confidence in the new data sample. If the new sample is noisy or $\sigma_N^2 \to \infty$, we do not correct the previous LSE. On the other hand, if the new sample is noise-free or $\sigma_N^2 \to 0$, then $\hat{A}[N] \to x[N]$. We discard all the previous samples. The gain factor then represents our confidence in the new data sample relative to the previous ones. We can make a further interpretation of our results if indeed $x[n] = A + w[n]$, where $w[n]$ is zero mean uncorrelated noise with variance σ_n^2. Then, we know from Chapter 6 (see Example 6.2) that the LSE is actually the BLUE, and therefore,

$$\text{var}(\hat{A}[N-1]) = \frac{1}{\sum_{n=0}^{N-1}\frac{1}{\sigma_n^2}}$$

and the gain factor for the Nth correction is (see (8.37))

$$\begin{aligned} K[N] &= \frac{\frac{1}{\sigma_N^2}}{\sum_{n=0}^{N}\frac{1}{\sigma_n^2}} \\ &= \frac{\frac{1}{\sigma_N^2}}{\frac{1}{\sigma_N^2} + \frac{1}{\text{var}(\hat{A}[N-1])}} \\ &= \frac{\text{var}(\hat{A}[N-1])}{\text{var}(\hat{A}[N-1]) + \sigma_N^2}. \end{aligned} \tag{8.38}$$

Since $0 \leq K[N] \leq 1$, the correction is large if $K[N]$ is large or $\text{var}(\hat{A}[N-1])$ is large. Likewise, if the variance of the previous estimator is small, then so is the correction. Further expressions may be developed for determining the gain recursively, since $K[N]$

depends on $\text{var}(\hat{A}[N-1])$. We can express the latter as

$$\begin{aligned}
\text{var}(\hat{A}[N]) &= \frac{1}{\sum_{n=0}^{N} \frac{1}{\sigma_n^2}} \\
&= \frac{1}{\sum_{n=0}^{N-1} \frac{1}{\sigma_n^2} + \frac{1}{\sigma_N^2}} \\
&= \frac{1}{\frac{1}{\text{var}(\hat{A}[N-1])} + \frac{1}{\sigma_N^2}} \\
&= \frac{\text{var}(\hat{A}[N-1])\sigma_N^2}{\text{var}(\hat{A}[N-1]) + \sigma_N^2} \\
&= \left(1 - \frac{\text{var}(\hat{A}[N-1])}{\text{var}(\hat{A}[N-1]) + \sigma_N^2}\right) \text{var}(\hat{A}[N-1])
\end{aligned}$$

or finally

$$\text{var}(\hat{A}[N]) = (1 - K[N])\text{var}(\hat{A}[N-1]). \tag{8.39}$$

To find $K[N]$ recursively we can use (8.38) and (8.39) as summarized below. In the process of recursively finding the gain, the variance of the LSE is also found recursively. Summarizing our results, we have

Estimator Update:

$$\hat{A}[N] = \hat{A}[N-1] + K[N](x[N] - \hat{A}[N-1]) \tag{8.40}$$

where

$$K[N] = \frac{\text{var}(\hat{A}[N-1])}{\text{var}(\hat{A}[N-1]) + \sigma_N^2}. \tag{8.41}$$

Variance Update:

$$\text{var}(\hat{A}[N]) = (1 - K[N])\text{var}(\hat{A}[N-1]). \tag{8.42}$$

To start the recursion we use

$$\begin{aligned}
\hat{A}[0] &= x[0] \\
\text{var}(\hat{A}[0]) &= \sigma_0^2.
\end{aligned}$$

Then, $K[1]$ is found from (8.41), and $\hat{A}[1]$ from (8.40). Next, $\text{var}(\hat{A}[1])$ is determined from (8.42). Continuing in this manner, we find $K[2]$, $\hat{A}[2]$, $\text{var}(\hat{A}[2])$, etc. An example of the sequential LS approach is shown in Figure 8.9 for $A = 10$, $\sigma_n^2 = 1$ using a Monte Carlo computer simulation. The variance and gain sequences have been computed

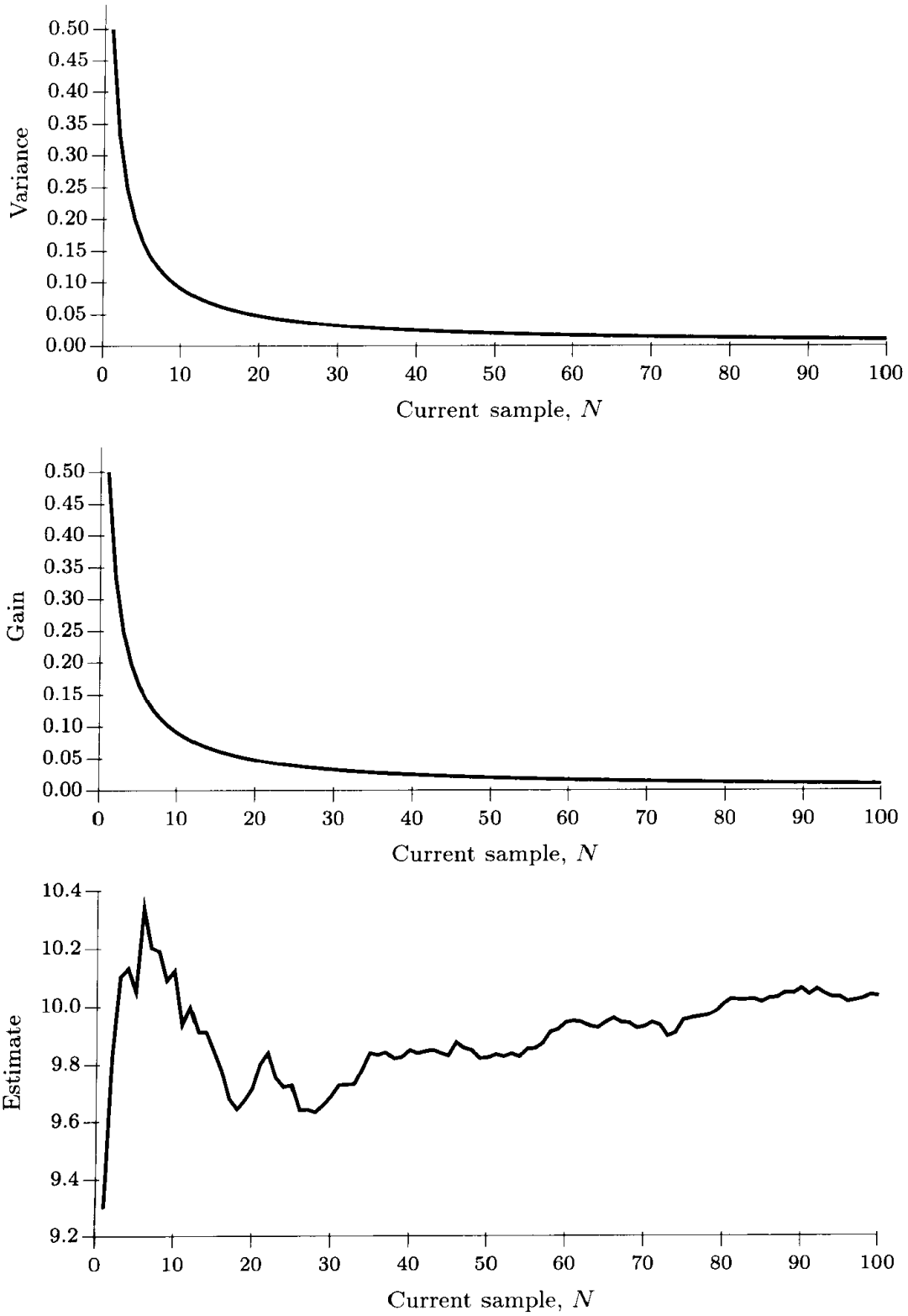


Figure 8.9 Sequential least squares for DC level in white Gaussian noise

recursively from (8.42) and (8.41), respectively. Note that they decrease to zero since

$$\text{var}(\hat{A}[N]) = \frac{1}{\sum_{n=0}^{N} \frac{1}{\sigma_n^2}} = \frac{1}{N+1}$$

and from (8.41)

$$K[N] = \frac{\frac{1}{N}}{\frac{1}{N}+1} = \frac{1}{N+1}.$$

Also, as seen in Figure 8.9c, the estimate appears to be converging to the true value of $A = 10$. This is in agreement with the variance approaching zero. Finally, it can be shown (see Problem 8.19) that the minimum LS error can be computed recursively as

$$J_{\min}[N] = J_{\min}[N-1] + \frac{\left(x[N] - \hat{A}[N-1]\right)^2}{\text{var}(\hat{A}[N-1]) + \sigma_N^2}. \tag{8.43}$$

We now generalize our results to obtain the sequential LSE for a vector parameter. The derivation is given in Appendix 8C. Consider the minimization of the weighted LS error criterion J with $\mathbf{W} = \mathbf{C}^{-1}$, where $\mathbf{C}$ denotes the covariance matrix of the zero mean noise or

$$J = (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T \mathbf{C}^{-1} (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}).$$

This is implicitly the same assumptions as in the BLUE. We know from (8.16) that the weighted LSE is

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{x}$$

and since $\mathbf{C}$ is the covariance matrix of the noise, we have from (6.17)

$$\mathbf{C}_{\hat{\theta}} = (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1}$$

where $\mathbf{C}_{\hat{\theta}}$ is the covariance matrix of $\hat{\boldsymbol{\theta}}$. *If* $\mathbf{C}$ *is diagonal or the noise is uncorrelated*, then $\hat{\boldsymbol{\theta}}$ may be computed sequentially in time, but otherwise not. Assuming this condition to hold, let

$$\begin{aligned} \mathbf{C}[n] &= \text{diag}(\sigma_0^2, \sigma_1^2, \ldots, \sigma_n^2) \\ \mathbf{H}[n] &= \begin{bmatrix} \mathbf{H}[n-1] \\ \mathbf{h}^T[n] \end{bmatrix} = \begin{bmatrix} n \times p \\ 1 \times p \end{bmatrix} \\ \mathbf{x}[n] &= \begin{bmatrix} x[0] & x[1] & \ldots & x[n] \end{bmatrix}^T \end{aligned}$$

and denote the weighted LSE of $\boldsymbol{\theta}$ based on $\mathbf{x}[n]$ or the $(n+1)$ data samples as $\hat{\boldsymbol{\theta}}[n]$. Then, the batch estimator is

$$\hat{\boldsymbol{\theta}}[n] = (\mathbf{H}^T[n] \mathbf{C}^{-1}[n] \mathbf{H}[n])^{-1} \mathbf{H}^T[n] \mathbf{C}^{-1}[n] \mathbf{x}[n] \tag{8.44}$$

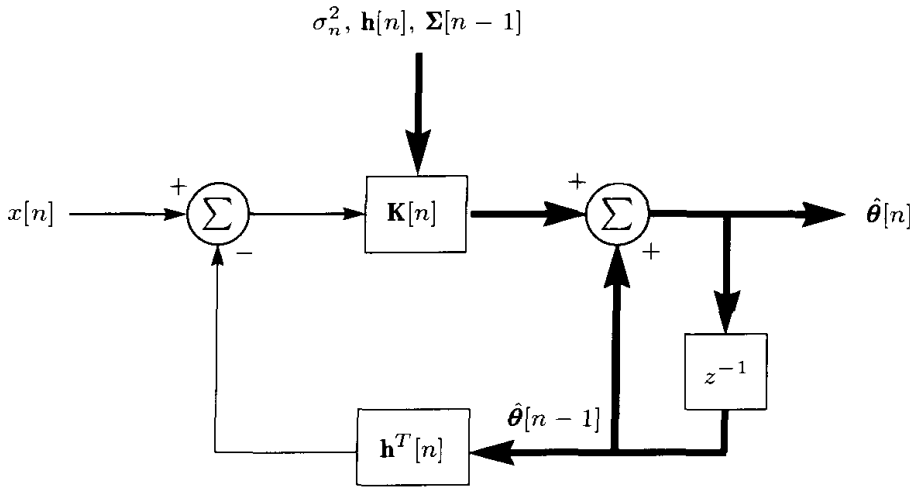


Figure 8.10 Sequential least squares estimator

with covariance matrix

$$\mathbf{C}_{\hat{\theta}} = \mathbf{\Sigma}[n] = (\mathbf{H}^T[n]\mathbf{C}^{-1}[n]\mathbf{H}[n])^{-1}. \tag{8.45}$$

It should be kept in mind that $\mathbf{C}[n]$ is the covariance matrix of the noise, while $\mathbf{\Sigma}[n]$ is the covariance matrix of the LSE. The sequential LSE becomes

Estimator Update:

$$\hat{\boldsymbol{\theta}}[n] = \hat{\boldsymbol{\theta}}[n-1] + \mathbf{K}[n]\left(x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1]\right) \tag{8.46}$$

where

$$\mathbf{K}[n] = \frac{\mathbf{\Sigma}[n-1]\mathbf{h}[n]}{\sigma_n^2 + \mathbf{h}^T[n]\mathbf{\Sigma}[n-1]\mathbf{h}[n]}. \tag{8.47}$$

Covariance Update:

$$\mathbf{\Sigma}[n] = (\mathbf{I} - \mathbf{K}[n]\mathbf{h}^T[n])\,\mathbf{\Sigma}[n-1]. \tag{8.48}$$

The gain factor $\mathbf{K}[n]$ is a $p \times 1$ vector, and the covariance matrix $\mathbf{\Sigma}[n]$ has dimension $p \times p$. It is of great interest that no matrix inversions are required. The estimator update is summarized in Figure 8.10, where the thick arrows indicate vector processing. To start the recursion we need to specify initial values for $\hat{\boldsymbol{\theta}}[n-1]$ and $\mathbf{\Sigma}[n-1]$, so that $\mathbf{K}[n]$ can be determined from (8.47) and then $\hat{\boldsymbol{\theta}}[n]$ from (8.46). In deriving (8.46)–(8.48) it was assumed that $\hat{\boldsymbol{\theta}}[n-1]$ and $\mathbf{\Sigma}[n-1]$ were available or that $\mathbf{H}^T[n-1]\mathbf{C}^{-1}[n-1]\mathbf{H}[n-1]$ was invertible as per (8.44) and (8.45). For this to be invertible $\mathbf{H}[n-1]$ must have rank greater than or equal to p. Since $\mathbf{H}[n-1]$ is an $n \times p$ matrix, we must have $n \geq p$ (assuming all its columns are linearly independent). Hence, the sequential LS procedure normally determines $\hat{\boldsymbol{\theta}}[p-1]$ and $\mathbf{\Sigma}[p-1]$ using the batch estimator (8.44) and (8.45), and then employs the sequential equations (8.46)–(8.48) for $n \geq p$. A second method of initializing the recursion is to assign values for $\hat{\boldsymbol{\theta}}[-1]$ and $\mathbf{\Sigma}[-1]$. Then, the sequential

LS estimator is computed for $n \geq 0$. This has the effect of biasing the estimator toward $\hat{\boldsymbol{\theta}}[-1]$. Typically, to minimize the biasing effect we choose $\boldsymbol{\Sigma}[-1]$ to be large (little confidence in $\hat{\boldsymbol{\theta}}[-1]$) or $\boldsymbol{\Sigma}[-1] = \alpha\mathbf{I}$, where α is large, and also $\hat{\boldsymbol{\theta}}[-1] = \mathbf{0}$. The LSE for $n \geq p$ will be the same as when the batch estimator is used for initialization if $\alpha \to \infty$ (see Problem 8.23). In the next example we show how to set up the equations. In Example 8.13 we apply this procedure to a signal processing problem.

Example 8.7 - Fourier Analysis

We now continue Example 8.4 in which the signal model is

$$s[n] = a\cos 2\pi f_0 n + b\sin 2\pi f_0 n \qquad n \geq 0$$

and $\boldsymbol{\theta} = [a\, b]^T$ is to be estimated by a sequential LSE. We furthermore assume that the noise is uncorrelated ($\mathbf{C}[n]$ *must be diagonal* for sequential LS to apply) and has equal variance σ^2 for each data sample. Since there are two parameters to be estimated, we need at least two observations or $x[0], x[1]$ to initialize the procedure using a batch initialization approach. We compute our first LSE, using (8.44), as

$$\begin{aligned}\hat{\boldsymbol{\theta}}[1] &= \left(\mathbf{H}^T[1]\left(\frac{1}{\sigma^2}\mathbf{I}\right)\mathbf{H}[1]\right)^{-1}\mathbf{H}^T[1]\left(\frac{1}{\sigma^2}\mathbf{I}\right)\mathbf{x}[1] \\ &= (\mathbf{H}^T[1]\mathbf{H}[1])^{-1}\mathbf{H}^T[1]\mathbf{x}[1]\end{aligned}$$

where

$$\begin{aligned}\mathbf{H}[1] &= \begin{bmatrix} 1 & 0 \\ \cos 2\pi f_0 & \sin 2\pi f_0 \end{bmatrix} \\ \mathbf{x}[1] &= \begin{bmatrix} x[0] \\ x[1] \end{bmatrix}.\end{aligned}$$

($\mathbf{H}[1]$ is the 2×2 partition of $\mathbf{H}$ given in (8.18).) The initial covariance matrix is, from (8.45),

$$\begin{aligned}\boldsymbol{\Sigma}[1] &= \left[\mathbf{H}^T[1]\left(\frac{1}{\sigma^2}\mathbf{I}\right)\mathbf{H}[1]\right]^{-1} \\ &= \sigma^2(\mathbf{H}^T[1]\mathbf{H}[1])^{-1}.\end{aligned}$$

Next we determine $\hat{\boldsymbol{\theta}}[2]$. To do so we first compute the 2×1 gain vector from (8.47) as

$$\mathbf{K}[2] = \frac{\boldsymbol{\Sigma}[1]\mathbf{h}[2]}{\sigma^2 + \mathbf{h}^T[2]\boldsymbol{\Sigma}[1]\mathbf{h}[2]}$$

where $\mathbf{h}^T[2]$ is the new row of the $\mathbf{H}[2]$ matrix or

$$\mathbf{h}^T[2] = \begin{bmatrix} \cos 4\pi f_0 & \sin 4\pi f_0 \end{bmatrix}.$$

Once the gain vector has been found, $\hat{\boldsymbol{\theta}}[2]$ is determined from (8.46) as

$$\hat{\boldsymbol{\theta}}[2] = \hat{\boldsymbol{\theta}}[1] + \mathbf{K}[2](x[2] - \mathbf{h}^T[2]\hat{\boldsymbol{\theta}}[1]).$$

Finally, the 2×2 LSE covariance matrix is updated as per (8.48) for use in the next computation of the gain vector or

$$\boldsymbol{\Sigma}[2] = (\mathbf{I} - \mathbf{K}[2]\mathbf{h}^T[2])\boldsymbol{\Sigma}[1].$$

It should be clear that, in general, computer evaluation of $\hat{\boldsymbol{\theta}}[n]$ is necessary. Also, except for the initialization procedure, no matrix inversion is required. Alternatively, we could have avoided even the matrix inversion of the initialization by assuming $\hat{\boldsymbol{\theta}}[-1] = \mathbf{0}$ and $\boldsymbol{\Sigma}[-1] = \alpha\mathbf{I}$ with α large. The recursion then would have begun at $n = 0$, and for $n \geq 2$ we would have the same result as before for large enough α. $\diamond$

Finally, we remark that if the minimum LS error is desired, it too can be found sequentially as (see Appendix 8C)

$$J_{\min}[n] = J_{\min}[n-1] + \frac{(x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1])^2}{\sigma_n^2 + \mathbf{h}^T[n]\boldsymbol{\Sigma}[n-1]\mathbf{h}[n]}. \tag{8.49}$$

8.8 Constrained Least Squares

At times we are confronted with LS problems whose unknown parameters must be constrained. Such would be the case if we wished to estimate the amplitudes of several signals but knew a priori that some of the amplitudes were equal. Then, the total number of parameters to be estimated should be reduced to take advantage of the a priori knowledge. For this example the parameters are linearly related, leading to a linear least squares problem with *linear constraints*. This problem is easily solved, as we now show.

Assume that the parameter $\boldsymbol{\theta}$ is subject to $r < p$ linear constraints. The constraints must be independent, ruling out the possibility of redundant constraints such as $\theta_1 + \theta_2 = 0$, $2\theta_1 + 2\theta_2 = 0$, where θ_i is the ith element of $\boldsymbol{\theta}$. We summarize the constraints as

$$\mathbf{A}\boldsymbol{\theta} = \mathbf{b} \tag{8.50}$$

where $\mathbf{A}$ is a known $r \times p$ matrix and $\mathbf{b}$ is a known $r \times 1$ vector. If, for instance, $p = 2$ and one parameter is known to be the negative of the other, then the constraint would be $\theta_1 + \theta_2 = 0$. We would then have $\mathbf{A} = [1\ 1]$ and $\mathbf{b} = \mathbf{0}$. It is always assumed that the matrix $\mathbf{A}$ is full rank (equal to r), which is necessary for the constraints to be independent. It should be realized that in the constrained LS problem there are actually only $(p - r)$ independent parameters.

To find the LSE subject to the linear constraints we use the technique of Lagrangian multipliers. We determine $\hat{\boldsymbol{\theta}}_c$ (c denotes the constrained LSE) by minimizing the Lagrangian

$$J_c = (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T(\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) + \boldsymbol{\lambda}^T(\mathbf{A}\boldsymbol{\theta} - \mathbf{b})$$

where $\boldsymbol{\lambda}$ is a $r \times 1$ vector of Lagrangian multipliers. Expanding this expression, we have

$$J_c = \mathbf{x}^T\mathbf{x} - 2\boldsymbol{\theta}^T\mathbf{H}^T\mathbf{x} + \boldsymbol{\theta}^T\mathbf{H}^T\mathbf{H}\boldsymbol{\theta} + \boldsymbol{\lambda}^T\mathbf{A}\boldsymbol{\theta} - \boldsymbol{\lambda}^T\mathbf{b}.$$

Taking the gradient with respect to $\boldsymbol{\theta}$ and using (4.3) yields

$$\frac{\partial J_c}{\partial \boldsymbol{\theta}} = -2\mathbf{H}^T\mathbf{x} + 2\mathbf{H}^T\mathbf{H}\boldsymbol{\theta} + \mathbf{A}^T\boldsymbol{\lambda}$$

and setting it equal to zero produces

$$\begin{aligned} \hat{\boldsymbol{\theta}}_c &= (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x} - \frac{1}{2}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{A}^T\boldsymbol{\lambda} \\ &= \hat{\boldsymbol{\theta}} - (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{A}^T\frac{\boldsymbol{\lambda}}{2} \end{aligned} \tag{8.51}$$

where $\hat{\boldsymbol{\theta}}$ is the *unconstrained* LSE and $\boldsymbol{\lambda}$ is yet to be determined. To find $\boldsymbol{\lambda}$ we impose the constraint of (8.50), so that

$$\mathbf{A}\boldsymbol{\theta}_c = \mathbf{A}\hat{\boldsymbol{\theta}} - \mathbf{A}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{A}^T\frac{\boldsymbol{\lambda}}{2} = \mathbf{b}$$

and hence

$$\frac{\boldsymbol{\lambda}}{2} = \left[\mathbf{A}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{A}^T\right]^{-1}(\mathbf{A}\hat{\boldsymbol{\theta}} - \mathbf{b}).$$

Substituting into (8.51) produces the solution

$$\hat{\boldsymbol{\theta}}_c = \hat{\boldsymbol{\theta}} - (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{A}^T\left[\mathbf{A}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{A}^T\right]^{-1}(\mathbf{A}\hat{\boldsymbol{\theta}} - \mathbf{b}) \tag{8.52}$$

where $\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$. The constrained LSE is a corrected version of the unconstrained LSE. If it happens that the constraint is fortuitously satisfied by $\hat{\boldsymbol{\theta}}$ or $\mathbf{A}\hat{\boldsymbol{\theta}} = \mathbf{b}$, then according to (8.52) the estimators are identical. Such is usually not the case, however. We consider a simple example.

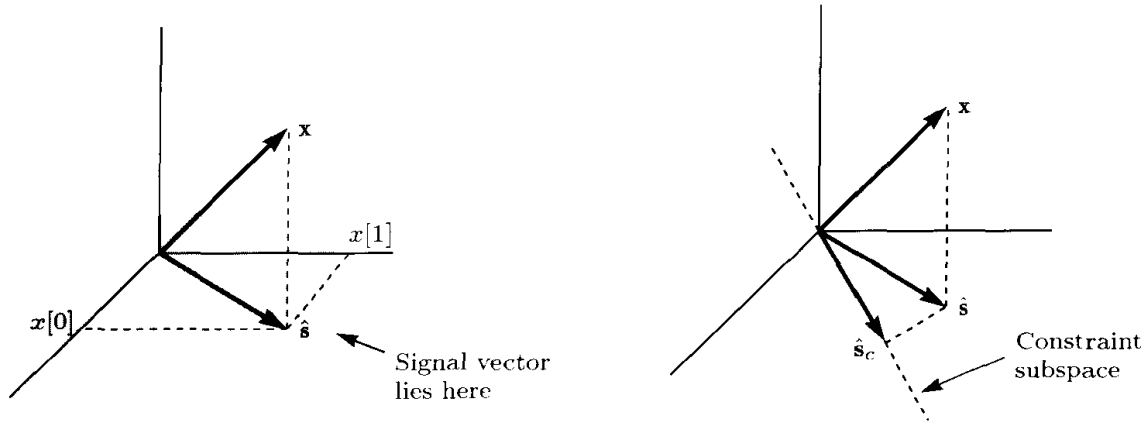
Example 8.8 - Constrained Signal

If the signal model is

$$s[n] = \begin{cases} \theta_1 & n = 0 \\ \theta_2 & n = 1 \\ 0 & n = 2 \end{cases}$$

and we observe $\{x[0], x[1], x[2]\}$, then the observation matrix is

$$\mathbf{H} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$



(a) Unconstrained least squares **(b)** Constrained least squares

Figure 8.11 Comparison of least squares and constrained least squares

Observe that the signal vector

$$\mathbf{s} = \mathbf{H}\boldsymbol{\theta} = \begin{bmatrix} \theta_1 \\ \theta_2 \\ 0 \end{bmatrix}$$

must lie in the plane shown in Figure 8.11a. The unconstrained LSE is

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x} = \begin{bmatrix} x[0] \\ x[1] \end{bmatrix}$$

and the signal estimate is

$$\hat{\mathbf{s}} = \mathbf{H}\hat{\boldsymbol{\theta}} = \begin{bmatrix} x[0] \\ x[1] \\ 0 \end{bmatrix}.$$

As shown in Figure 8.11a, this is intuitively reasonable. Now assume that we know a priori that $\theta_1 = \theta_2$. In terms of (8.50) we have

$$\begin{bmatrix} 1 & -1 \end{bmatrix} \boldsymbol{\theta} = 0$$

so that $\mathbf{A} = [1 \ -1]$ and $\mathbf{b} = 0$. Noting that $\mathbf{H}^T\mathbf{H} = \mathbf{I}$, we have from (8.52)

$$\begin{aligned} \hat{\boldsymbol{\theta}}_c &= \hat{\boldsymbol{\theta}} - \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{A}\hat{\boldsymbol{\theta}} \\ &= \left[\mathbf{I} - \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}\mathbf{A}\right]\hat{\boldsymbol{\theta}}. \end{aligned}$$

After some simple algebra this becomes

$$\hat{\boldsymbol{\theta}}_c = \begin{bmatrix} \frac{1}{2}(x[0] + x[1]) \\ \frac{1}{2}(x[0] + x[1]) \end{bmatrix}$$

and the constrained signal estimate becomes

$$\hat{\mathbf{s}}_c = \mathbf{H}\hat{\boldsymbol{\theta}}_c = \begin{bmatrix} \frac{1}{2}(x[0]+x[1]) \\ \frac{1}{2}(x[0]+x[1]) \\ 0 \end{bmatrix}$$

as illustrated in Figure 8.11b. Since $\theta_1 = \theta_2$, we just average the two observations, which is again intuitively reasonable. In this simple problem we could have just as easily incorporated our parameter constraints into the signal model directly to yield

$$s[n] = \begin{cases} \theta & n = 0 \\ \theta & n = 1 \\ 0 & n = 2 \end{cases}$$

and estimated θ. This would have produced the same result, as it must. This new model is sometimes referred to as the *reduced* model [Graybill 1976] for obvious reasons. It is worthwhile to view the constrained LS problem geometrically. Again referring to Figure 8.11b, if $\theta_1 = \theta_2$, then the signal model is

$$\mathbf{s} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \theta$$

and $\mathbf{s}$ must lie in the constraint subspace shown. The constrained signal estimate $\hat{\mathbf{s}}_c$ may be viewed as the projection of the unconstrained signal estimate $\hat{\mathbf{s}}$ onto the constrained subspace. This accounts for the correction term of (8.52). In fact, (8.52) may be obtained geometrically using projection theory. See also Problem 8.24. ◇

8.9 Nonlinear Least Squares

In Section 8.3 we introduced the nonlinear LS problem. We now investigate this in more detail. Recall that the LS procedure estimates model parameters $\boldsymbol{\theta}$ by minimizing the LS error criterion

$$J = (\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}))^T(\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}))$$

where $\mathbf{s}(\boldsymbol{\theta})$ is the signal model for $\mathbf{x}$, with its dependence on $\boldsymbol{\theta}$ explicitly shown. (Note that if $\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}) \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$, the LSE is also the MLE.) In the linear LS problem the signal takes on the special form $\mathbf{s}(\boldsymbol{\theta}) = \mathbf{H}\boldsymbol{\theta}$, which leads to the simple linear LSE. In general, $\mathbf{s}(\boldsymbol{\theta})$ cannot be expressed in this manner but is an N-dimensional nonlinear function of $\boldsymbol{\theta}$. In such a case the minimization of J becomes much more difficult, if not impossible. This type of nonlinear LS problem is often termed a *nonlinear regression problem* in statistics [Bard 1974, Seber and Wild 1989], and much theoretical work on it can be found. Practically, the determination of the nonlinear LSE must be based on

iterative approaches and so suffers from the same limitations discussed in Chapter 7 for MLE determination by numerical methods. The preferable method of a grid search is practical only if the dimensionality of $\boldsymbol{\theta}$ is small, perhaps $p \leq 5$.

Before discussing general methods for determining nonlinear LSEs we first describe two methods that can reduce the complexity of the problem. They are

1. transformation of parameters
2. separability of parameters.

In the first case we seek a one-to-one transformation of $\boldsymbol{\theta}$ that produces a *linear* signal model in the new space. To do so we let

$$\boldsymbol{\alpha} = \mathbf{g}(\boldsymbol{\theta})$$

where $\mathbf{g}$ is a p-dimensional function of $\boldsymbol{\theta}$ whose inverse exists. If a $\mathbf{g}$ can be found so that

$$\mathbf{s}(\boldsymbol{\theta}(\boldsymbol{\alpha})) = \mathbf{s}\left(\mathbf{g}^{-1}(\boldsymbol{\alpha})\right) = \mathbf{H}\boldsymbol{\alpha}$$

then the signal model will be linear in $\boldsymbol{\alpha}$. We can then easily find the *linear* LSE of $\boldsymbol{\alpha}$ and thus the *nonlinear* LSE of $\boldsymbol{\theta}$ by

$$\hat{\boldsymbol{\theta}} = \mathbf{g}^{-1}(\hat{\boldsymbol{\alpha}})$$

where

$$\hat{\boldsymbol{\alpha}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}.$$

This approach relies on the property that the minimization can be carried out in any transformed space that is obtained by a one-to-one mapping and then converted back to the original space (see Problem 8.26). The determination of the transformation $\mathbf{g}$, if it exists, is usually quite difficult. Suffice it to say, only a few nonlinear LS problems may be solved in this manner.

Example 8.9 - Sinusoidal Parameter Estimation

For a sinusoidal signal model

$$s[n] = A\cos(2\pi f_0 n + \phi) \qquad n = 0, 1, \ldots, N-1$$

it is desired to estimate the amplitude A, where $A > 0$, and phase ϕ. The frequency f_0 is assumed known. The LSE is obtained by minimizing

$$J = \sum_{n=0}^{N-1} \left(x[n] - A\cos(2\pi f_0 n + \phi)\right)^2$$

over A and ϕ, a nonlinear LS problem. However, because

$$A\cos(2\pi f_0 n + \phi) = A\cos\phi\cos 2\pi f_0 n - A\sin\phi\sin 2\pi f_0 n$$

if we let

$$\begin{aligned} \alpha_1 &= A\cos\phi \\ \alpha_2 &= -A\sin\phi, \end{aligned}$$

then the signal model becomes

$$s[n] = \alpha_1 \cos 2\pi f_0 n + \alpha_2 \sin 2\pi f_0 n.$$

In matrix form this is

$$\mathbf{s} = \mathbf{H}\boldsymbol{\alpha}$$

where

$$\mathbf{H} = \begin{bmatrix} 1 & 0 \\ \cos 2\pi f_0 & \sin 2\pi f_0 \\ \vdots & \vdots \\ \cos 2\pi f_0(N-1) & \sin 2\pi f_0(N-1) \end{bmatrix},$$

which is now linear in the new parameters. The LSE of $\boldsymbol{\alpha}$ is

$$\hat{\boldsymbol{\alpha}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$$

and to find $\hat{\boldsymbol{\theta}}$ we must find the inverse transformation $\mathbf{g}^{-1}(\boldsymbol{\alpha})$. This is

$$\begin{aligned} A &= \sqrt{\alpha_1^2 + \alpha_2^2} \\ \phi &= \arctan\left(\frac{-\alpha_2}{\alpha_1}\right) \end{aligned}$$

so that the nonlinear LSE for this problem is given by

$$\begin{aligned} \hat{\boldsymbol{\theta}} &= \begin{bmatrix} \hat{A} \\ \hat{\phi} \end{bmatrix} \\ &= \begin{bmatrix} \sqrt{\hat{\alpha}_1^2 + \hat{\alpha}_2^2} \\ \arctan\left(\frac{-\hat{\alpha}_2}{\hat{\alpha}_1}\right) \end{bmatrix} \end{aligned}$$

where $\hat{\boldsymbol{\alpha}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$. The reader may wish to refer back to Example 7.16 in which this approach was used to find the MLE of frequency. ◇

A second type of nonlinear LS problem that is less complex than the general one exhibits the separability property. Although the signal model is nonlinear, it may be linear in some of the parameters, as illustrated in Example 8.3. In general, a separable signal model has the form

$$\mathbf{s} = \mathbf{H}(\boldsymbol{\alpha})\boldsymbol{\beta}$$

where

$$\boldsymbol{\theta} = \begin{bmatrix} \boldsymbol{\alpha} \\ \boldsymbol{\beta} \end{bmatrix} = \begin{bmatrix} (p-q) \times 1 \\ q \times 1 \end{bmatrix}$$

and $\mathbf{H}(\boldsymbol{\alpha})$ is an $N \times q$ matrix dependent on $\boldsymbol{\alpha}$. This model is linear in $\boldsymbol{\beta}$ but nonlinear in $\boldsymbol{\alpha}$. As a result, the LS error may be minimized with respect to $\boldsymbol{\beta}$ and thus reduced to a function of $\boldsymbol{\alpha}$ only. Since

$$J(\boldsymbol{\alpha}, \boldsymbol{\beta}) = (\mathbf{x} - \mathbf{H}(\boldsymbol{\alpha})\boldsymbol{\beta})^T (\mathbf{x} - \mathbf{H}(\boldsymbol{\alpha})\boldsymbol{\beta})$$

the $\boldsymbol{\beta}$ that minimizes J for a given $\boldsymbol{\alpha}$ is

$$\hat{\boldsymbol{\beta}} = \left(\mathbf{H}^T(\boldsymbol{\alpha})\mathbf{H}(\boldsymbol{\alpha})\right)^{-1} \mathbf{H}^T(\boldsymbol{\alpha})\mathbf{x} \tag{8.53}$$

and the resulting LS error is, from (8.22),

$$J(\boldsymbol{\alpha}, \hat{\boldsymbol{\beta}}) = \mathbf{x}^T \left[\mathbf{I} - \mathbf{H}(\boldsymbol{\alpha}) \left(\mathbf{H}^T(\boldsymbol{\alpha})\mathbf{H}(\boldsymbol{\alpha})\right)^{-1} \mathbf{H}^T(\boldsymbol{\alpha})\right] \mathbf{x}.$$

The problem now reduces to a *maximization* of

$$\mathbf{x}^T\mathbf{H}(\boldsymbol{\alpha}) \left(\mathbf{H}^T(\boldsymbol{\alpha})\mathbf{H}(\boldsymbol{\alpha})\right)^{-1} \mathbf{H}^T(\boldsymbol{\alpha})\mathbf{x} \tag{8.54}$$

over $\boldsymbol{\alpha}$. If, for instance, $q = p - 1$, so that $\boldsymbol{\alpha}$ is a scalar, then a grid search can possibly be used. This should be contrasted with the original minimization of a p-dimensional function. (See also Example 7.16.)

Example 8.10 - Damped Exponentials

Assume we have a signal model of the form

$$s[n] = A_1 r^n + A_2 r^{2n} + A_3 r^{3n}$$

where the unknown parameters are $\{A_1, A_2, A_3, r\}$. It is known that $0 < r < 1$. Then, the model is linear in the amplitudes $\boldsymbol{\beta} = [A_1\ A_2\ A_3]^T$, and nonlinear in the damping factor $\boldsymbol{\alpha} = r$. Using (8.54), the nonlinear LSE is obtained by maximizing

$$\mathbf{x}^T\mathbf{H}(r) \left(\mathbf{H}^T(r)\mathbf{H}(r)\right)^{-1} \mathbf{H}^T(r)\mathbf{x}$$

over $0 < r < 1$, where

$$\mathbf{H}(r) = \begin{bmatrix} 1 & 1 & 1 \\ r & r^2 & r^3 \\ \vdots & \vdots & \vdots \\ r^{N-1} & r^{2(N-1)} & r^{3(N-1)} \end{bmatrix}.$$

Once $\hat{r}$ is found we have the LSE for the amplitudes

$$\hat{\boldsymbol{\beta}} = \left(\mathbf{H}^T(\hat{r})\mathbf{H}(\hat{r})\right)^{-1} \mathbf{H}^T(\hat{r})\mathbf{x}.$$

This maximization is easily carried out on a digital computer. ◇

It may also be possible to combine the approaches to reduce a nonseparable problem to a separable one by an appropriate transformation. When these approaches fail, we are forced to tackle the original nonlinear LS problem or to minimize

$$J = (\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}))^T (\mathbf{x} - \mathbf{s}(\boldsymbol{\theta})).$$

Necessary conditions can be obtained by differentiation of J. They are

$$\frac{\partial J}{\partial \theta_j} = -2 \sum_{i=0}^{N-1} (x[i] - s[i]) \frac{\partial s[i]}{\partial \theta_j} = 0$$

for $j = 1, 2, \ldots, p$. If we define an $N \times p$ Jacobian matrix as

$$\left[\frac{\partial \mathbf{s}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right]_{ij} = \frac{\partial s[i]}{\partial \theta_j} \qquad \begin{array}{l} i = 0, 1, \ldots, N-1 \\ j = 1, 2, \ldots, p, \end{array}$$

then the necessary conditions become

$$\sum_{i=0}^{N-1} (x[i] - s[i]) \left[\frac{\partial \mathbf{s}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right]_{ij} = 0 \qquad j = 1, 2, \ldots, p$$

or in matrix form

$$\frac{\partial \mathbf{s}(\boldsymbol{\theta})^T}{\partial \boldsymbol{\theta}} (\mathbf{x} - \mathbf{s}(\boldsymbol{\theta})) = \mathbf{0} \tag{8.55}$$

which is a set of p simultaneous nonlinear equations. (If the signal is linear in the unknown parameters or $\mathbf{s}(\boldsymbol{\theta}) = \mathbf{H}\boldsymbol{\theta}$ so that $\partial s(\boldsymbol{\theta})/\partial \boldsymbol{\theta} = \mathbf{H}$, we have our usual LS equations.) They may be solved using the Newton-Raphson iteration described in Chapter 7. Letting

$$\mathbf{g}(\boldsymbol{\theta}) = \frac{\partial \mathbf{s}(\boldsymbol{\theta})^T}{\partial \boldsymbol{\theta}} (\mathbf{x} - \mathbf{s}(\boldsymbol{\theta})) \tag{8.56}$$

the iteration becomes

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \left(\frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right)^{-1} \mathbf{g}(\boldsymbol{\theta}) \Big|_{\boldsymbol{\theta} = \boldsymbol{\theta}_k}. \tag{8.57}$$

To find the Jacobian of $\mathbf{g}$, which is actually the scaled Hessian of J, we need

$$\frac{\partial [\mathbf{g}(\boldsymbol{\theta})]_i}{\partial \theta_j} = \frac{\partial}{\partial \theta_j} \left[\sum_{n=0}^{N-1} (x[n] - s[n]) \frac{\partial s[n]}{\partial \theta_i}\right]$$

which follows from (8.56). Evaluating this, we have

$$\frac{\partial [\mathbf{g}(\boldsymbol{\theta})]_i}{\partial \theta_j} = \sum_{n=0}^{N-1} \left[(x[n] - s[n]) \frac{\partial^2 s[n]}{\partial \theta_i \partial \theta_j} - \frac{\partial s[n]}{\partial \theta_j} \frac{\partial s[n]}{\partial \theta_i}\right].$$

To put this in more succinct form let

$$[\mathbf{H}(\boldsymbol{\theta})]_{ij} = \left[\frac{\partial s(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right]_{ij} = \frac{\partial s[i]}{\partial \theta_j} \tag{8.58}$$

for $i = 0, 1, \ldots, N-1; j = 1, 2, \ldots, p$. (Again if $\mathbf{s}(\boldsymbol{\theta}) = \mathbf{H}\boldsymbol{\theta}$, then $\mathbf{H}(\boldsymbol{\theta}) = \mathbf{H}$.) Also let

$$[\mathbf{G}_n(\boldsymbol{\theta})]_{ij} = \frac{\partial^2 s[n]}{\partial \theta_i \partial \theta_j} \tag{8.59}$$

for $i = 1, 2, \ldots, p$; $j = 1, 2, \ldots, p$. Then,

$$\begin{aligned}\frac{\partial [\mathbf{g}(\boldsymbol{\theta})]_i}{\partial \theta_j} &= \sum_{n=0}^{N-1} (x[n] - s[n]) [\mathbf{G}_n(\boldsymbol{\theta})]_{ij} - [\mathbf{H}(\boldsymbol{\theta})]_{nj} [\mathbf{H}(\boldsymbol{\theta})]_{ni} \\ &= \sum_{n=0}^{N-1} [\mathbf{G}_n(\boldsymbol{\theta})]_{ij} (x[n] - s[n]) - [\mathbf{H}^T(\boldsymbol{\theta})]_{in} [\mathbf{H}(\boldsymbol{\theta})]_{nj}\end{aligned}$$

and

$$\frac{\partial \mathbf{g}(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \sum_{n=0}^{N-1} \mathbf{G}_n(\boldsymbol{\theta})(x[n] - s[n]) - \mathbf{H}^T(\boldsymbol{\theta})\mathbf{H}(\boldsymbol{\theta}). \tag{8.60}$$

In summary, we have upon using (8.57) with (8.56) and (8.60) the Newton-Raphson iteration

$$\begin{aligned}\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + &\left(\mathbf{H}^T(\boldsymbol{\theta}_k)\mathbf{H}(\boldsymbol{\theta}_k) - \sum_{n=0}^{N-1} \mathbf{G}_n(\boldsymbol{\theta}_k)(x[n] - [\mathbf{s}(\boldsymbol{\theta}_k)]_n)\right)^{-1} \\ &\cdot \mathbf{H}^T(\boldsymbol{\theta}_k)(\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}_k))\end{aligned} \tag{8.61}$$

where $\mathbf{H}(\boldsymbol{\theta})$ is given by (8.58) and $\mathbf{G}_n(\boldsymbol{\theta})$ is given by (8.59). These are the first-order and second-order partials of the signal with respect to the unknown parameters. It is interesting to note that if $\mathbf{s}(\boldsymbol{\theta}) = \mathbf{H}\boldsymbol{\theta}$, then $\mathbf{G}_n(\boldsymbol{\theta}) = \mathbf{0}$ and $\mathbf{H}(\boldsymbol{\theta}) = \mathbf{H}$ and we have

$$\begin{aligned}\boldsymbol{\theta}_{k+1} &= \boldsymbol{\theta}_k + (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T(\mathbf{x} - \mathbf{H}\boldsymbol{\theta}_k) \\ &= (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}\end{aligned}$$

or convergence is attained in one step. This is, of course, due to the exact quadratic nature of the LS error criterion, which results in a linear $\mathbf{g}(\boldsymbol{\theta})$. It would be expected then that for signal models that are approximately linear, convergence will occur rapidly.

A second method for solving a nonlinear LS problem is to linearize the signal model about some nominal $\boldsymbol{\theta}$ and then apply the linear LS procedure. This differs from the Newton-Raphson method in which the *derivative* of J is linearized about the current iterate. To appreciate the difference consider a scalar parameter and let θ_0 be the nominal value of θ. Then, linearizing about θ_0 we have

$$s[n;\theta] \approx s[n;\theta_0] + \left.\frac{\partial s[n;\theta]}{\partial \theta}\right|_{\theta=\theta_0} (\theta - \theta_0)$$

where the dependence of $s[n]$ on θ is now shown. The LS error becomes

$$\begin{aligned} J &= \sum_{n=0}^{N-1} (x[n] - s[n;\theta])^2 \\ &\approx \sum_{n=0}^{N-1} \left(x[n] - s[n;\theta_0] + \left.\frac{\partial s[n;\theta]}{\partial \theta}\right|_{\theta=\theta_0} \theta_0 - \left.\frac{\partial s[n;\theta]}{\partial \theta}\right|_{\theta=\theta_0} \theta \right)^2 \\ &= (\mathbf{x} - \mathbf{s}(\theta_0) + \mathbf{H}(\theta_0)\theta_0 - \mathbf{H}(\theta_0)\theta)^T (\mathbf{x} - \mathbf{s}(\theta_0) + \mathbf{H}(\theta_0)\theta_0 - \mathbf{H}(\theta_0)\theta) . \end{aligned}$$

Since $\mathbf{x} - \mathbf{s}(\theta_0) + \mathbf{H}(\theta_0)\theta_0$ is known, we have as the LSE

$$\begin{aligned} \hat{\theta} &= \left(\mathbf{H}^T(\theta_0)\mathbf{H}(\theta_0)\right)^{-1} \mathbf{H}^T(\theta_0) (\mathbf{x} - \mathbf{s}(\theta_0) + \mathbf{H}(\theta_0)\theta_0) \\ &= \theta_0 + \left(\mathbf{H}^T(\theta_0)\mathbf{H}(\theta_0)\right)^{-1} \mathbf{H}^T(\theta_0) (\mathbf{x} - \mathbf{s}(\theta_0)) . \end{aligned}$$

If we now iterate the solution, it becomes

$$\theta_{k+1} = \theta_k + \left(\mathbf{H}^T(\theta_k)\mathbf{H}(\theta_k)\right)^{-1} \mathbf{H}^T(\theta_k)(\mathbf{x} - \mathbf{s}(\theta_k))$$

which is identical to the Newton-Raphson iteration except for the omission of the second derivatives or for the presence of $\mathbf{G}_n$. The linearization method is termed the *Gauss-Newton method* and can easily be generalized to the vector parameter case as

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k + \left(\mathbf{H}^T(\boldsymbol{\theta}_k)\mathbf{H}(\boldsymbol{\theta}_k)\right)^{-1} \mathbf{H}^T(\boldsymbol{\theta}_k) (\mathbf{x} - \mathbf{s}(\boldsymbol{\theta}_k)) \tag{8.62}$$

where

$$[\mathbf{H}(\boldsymbol{\theta})]_{ij} = \frac{\partial s[i]}{\partial \theta_j}.$$

In Example 8.14 the Gauss method is illustrated. Both the Newton-Raphson and the Gauss methods can have convergence problems. It has been argued that neither method is reliable enough to use without safeguards. The interested reader should consult [Seber and Wild 1989] for additional implementation details.

8.10 Signal Processing Examples

We now describe some typical signal processing problems for which a LSE is used. In these applications the optimal MVU estimator is unavailable. The statistical characterization of the noise may not be known or, even if it is, the optimal MVU estimator cannot be found. For known noise statistics the asymptotically optimal MLE is generally too complicated to implement. Faced with these practical difficulties we resort to least squares.

Example 8.11 - Digital Filter Design

A common problem in digital signal processing is to design a digital filter whose frequency response closely matches a given frequency response specification [Oppenheim and Schafer 1975]. Alternatively, we could attempt to match the impulse response, which is the approach examined in this example. A general infinite impulse response (IIR) filter has the system function

$$\begin{aligned}\mathcal{H}(z) &= \frac{\mathcal{B}(z)}{\mathcal{A}(z)} \\ &= \frac{b[0] + b[1]z^{-1} + \cdots + b[q]z^{-q}}{1 + a[1]z^{-1} + \cdots + a[p]z^{-p}}.\end{aligned}$$

If the desired frequency response is $H_d(f) = \mathcal{H}_d(\exp[j2\pi f])$, then the inverse Fourier transform is the desired impulse response or

$$h_d[n] = \mathcal{F}^{-1}\{H_d(f)\}.$$

The digital filter impulse response is given as a recursive difference equation, obtained by taking the inverse z transform of $\mathcal{H}(z)$ to produce

$$h[n] = \begin{cases} -\sum_{k=1}^{p} a[k]h[n-k] + \sum_{k=0}^{q} b[k]\delta[n-k] & n \geq 0 \\ 0 & n < 0. \end{cases}$$

A straightforward LS solution would seek to choose $\{a[k], b[k]\}$ to minimize

$$J = \sum_{n=0}^{N-1} (h_d[n] - h[n])^2$$

where N is some sufficiently large integer for which $h_d[n]$ is essentially zero. Unfortunately, this approach produces a nonlinear LS problem (see Problem 8.28 for the exact solution). (The reader should note that $h_d[n]$ plays the role of the "data," and $h[n]$ that of the "signal" model. Also, the "noise" in the data is attributable to modeling error whose statistical characterization is unknown.) As an example, if

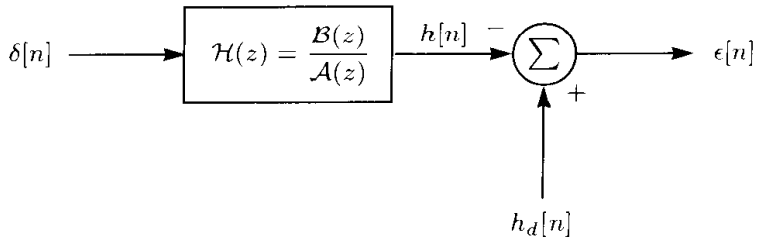
$$\mathcal{H}(z) = \frac{b[0]}{1 + a[1]z^{-1}},$$

then

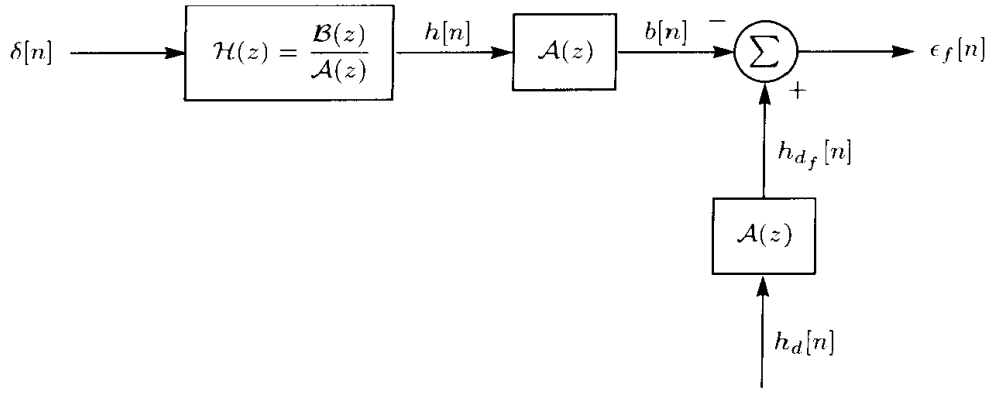
$$h[n] = \begin{cases} b[0](-a[1])^n & n \geq 0 \\ 0 & n < 0 \end{cases}$$

and

$$J = \sum_{n=0}^{N-1} (h_d[n] - b[0](-a[1])^n)^2$$



(a) True least squares error



(b) Filtered least squares error

Figure 8.12 Conversion of nonlinear least squares filter design to linear least squares

which is clearly very nonlinear in $a[1]$. In fact, it is the presence of $\mathcal{A}(z)$ that causes the LS error to be nonquadratic. To alleviate this problem we can filter both $h_d[n]$ and $h[n]$ by $\mathcal{A}(z)$, as shown in Figure 8.12. Then, we minimize the *filtered* LS error

$$J_f = \sum_{n=0}^{N-1} (h_{d_f}[n] - b[n])^2$$

where $h_{d_f}[n]$ is given as

$$h_{d_f}[n] = \sum_{k=0}^{p} a[k] h_d[n-k]$$

and $a[0] = 1$. The filtered LS error then becomes

$$J_f = \sum_{n=0}^{N-1} \left(\sum_{k=0}^{p} a[k] h_d[n-k] - b[n] \right)^2$$

which is now a quadratic function of the $a[k]$'s and $b[k]$'s. Alternatively,

$$J_f = \sum_{n=0}^{N-1} \left[h_d[n] - \left(-\sum_{k=1}^{p} a[k] h_d[n-k] + b[n] \right) \right]^2 .$$

In minimizing this over the filter coefficients note that the $b[n]$'s appear only in the first $(q+1)$ terms since $b[n] = 0$ for $n > q$. As a result, we have upon letting $\mathbf{a} = [a[1]\, a[2] \ldots a[p]]^T$ and $\mathbf{b} = [b[0]\, b[1] \ldots b[q]]^T$,

$$\begin{aligned} J_f(\mathbf{a}, \mathbf{b}) &= \sum_{n=0}^{q} \left[h_d[n] - \left(-\sum_{k=1}^{p} a[k] h_d[n-k] + b[n] \right) \right]^2 \\ &\quad + \sum_{n=q+1}^{N-1} \left[h_d[n] - \left(-\sum_{k=1}^{p} a[k] h_d[n-k] \right) \right]^2 . \end{aligned}$$

The first sum may be minimized (actually made to be zero) by setting

$$h_d[n] - \left(-\sum_{k=1}^{p} a[k] h_d[n-k] + b[n] \right) = 0 \qquad n = 0, 1, \ldots, q$$

or

$$b[n] = h_d[n] + \sum_{k=1}^{p} a[k] h_d[n-k].$$

In matrix form the LSE of the numerator coefficients is

$$\hat{\mathbf{b}} = \mathbf{h} + \mathbf{H}_0 \hat{\mathbf{a}}$$

where

$$\begin{aligned} \mathbf{h} &= \begin{bmatrix} h_d[0] & h_d[1] & \ldots & h_d[q] \end{bmatrix}^T \\ \mathbf{H}_0 &= \begin{bmatrix} 0 & 0 & \ldots & 0 \\ h_d[0] & 0 & \ldots & 0 \\ h_d[1] & h_d[0] & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots \\ h_d[q-1] & h_d[q-2] & \ldots & h_d[q-p] \end{bmatrix} . \end{aligned}$$

The vector $\mathbf{h}$ has dimension $(q+1) \times 1$, while the matrix $\mathbf{H}_0$ has dimension $(q+1) \times p$. To find the LSE of the denominator coefficients or $\hat{\mathbf{a}}$ we must minimize

$$\begin{aligned} J_f(\mathbf{a}, \hat{\mathbf{b}}) &= \sum_{n=q+1}^{N-1} \left[h_d[n] - \left(-\sum_{k=1}^{p} a[k] h_d[n-k] \right) \right]^2 \\ &= (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) \end{aligned}$$

where $\boldsymbol{\theta} = \mathbf{a}$ and

$$\mathbf{x} = \begin{bmatrix} h_d[q+1] & h_d[q+2] & \dots & h_d[N-1] \end{bmatrix}^T \qquad (N-1-q \times 1)$$

$$\mathbf{H} = -\begin{bmatrix} h_d[q] & h_d[q-1] & \dots & h_d[q-p+1] \\ h_d[q+1] & h_d[q] & \dots & h_d[q-p+2] \\ \vdots & \vdots & \vdots & \vdots \\ h_d[N-2] & h_d[N-3] & \dots & h_d[N-1-p] \end{bmatrix} \qquad (N-1-q \times p).$$

The LSE of the denominator coefficients is

$$\hat{\mathbf{a}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}.$$

This method for designing digital filters is termed the *least squares Prony method* [Parks and Burrus 1987]. As an example, consider the design of a low-pass filter for which the desired frequency response is

$$H_d'(f) = \begin{cases} 1 & |f| < f_c \\ 0 & |f| > f_c \end{cases}$$

where f_c is the cutoff frequency. The corresponding impulse response is

$$h_d'[n] = \frac{\sin 2\pi f_c n}{\pi n} \qquad -\infty < n < \infty$$

and as expected is not causal (due to the zero phase frequency response assumption). To ensure causality we delay the impulse response by n_0 samples and then set the resultant impulse response equal to zero for $n < 0$. Next, to approximate the desired impulse response using the LS Prony method we assume that N samples are available or

$$h_d[n] = \frac{\sin 2\pi f_c(n-n_0)}{\pi(n-n_0)} \qquad n = 0, 1, \dots, N-1.$$

For a cutoff frequency of $f_c = 0.1$ and a delay of $n_0 = 25$ samples, the desired impulse response is shown in Figure 8.13a for $N = 50$. The corresponding desired magnitude frequency response is shown in Figure 8.13b. The effects of truncating the impulse response are manifested by the sidelobe structure of the response. Using the LS Prony method with $p = q = 10$, a digital filter was designed to match the desired low-pass filter frequency response. The result is shown in Figure 8.13c, where the magnitude of the frequency response or $|\mathcal{H}(\exp(j2\pi f))|$ has been plotted in dB. The agreement is generally good, with the Prony digital filter exhibiting a peaky structure in the passband (frequencies below the cutoff) and a smooth rolloff in the stopband (frequencies above the cutoff) in contrast to the desired response. Larger values for p and q would presumably improve the match. ◇

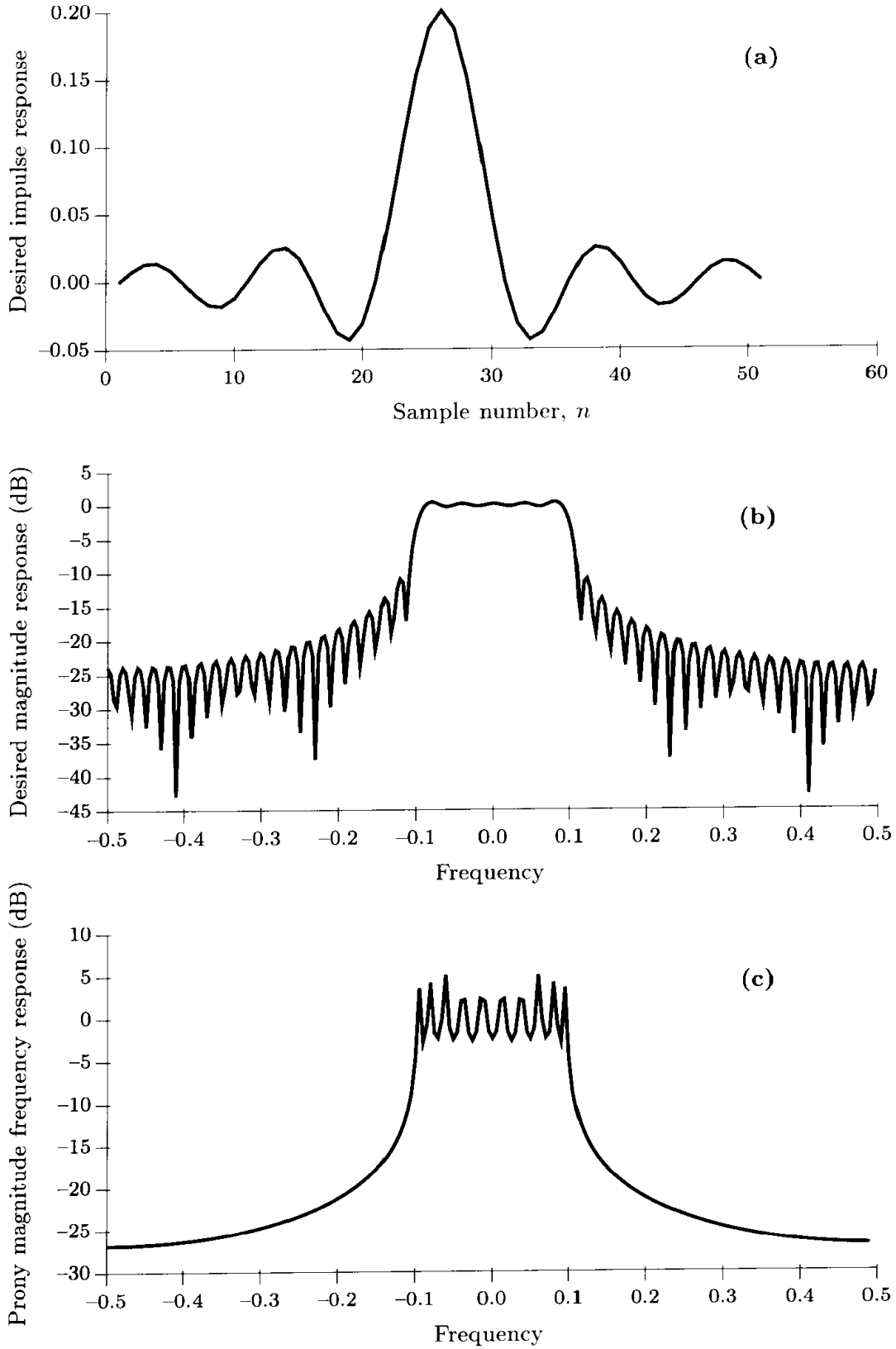


Figure 8.13 Low pass filter design by least squares Prony method

Example 8.12 - AR Parameter Estimation for the ARMA Model

We now describe a method for estimation of the AR parameters of an autoregressive moving average (ARMA) model. An ARMA model for a WSS random process assumes a PSD of

$$P_{xx}(f) = \frac{\sigma_u^2 |B(f)|^2}{|A(f)|^2}$$

where

$$\begin{aligned} B(f) &= 1 + \sum_{k=1}^{q} b[k] \exp(-j2\pi fk) \\ A(f) &= 1 + \sum_{k=1}^{p} a[k] \exp(-j2\pi fk). \end{aligned}$$

The $b[k]$'s are termed the MA filter parameters, the $a[k]$'s the AR filter parameters, and σ_u^2 the driving white noise variance. The process is obtained by exciting a *causal* filter whose frequency response is $B(f)/A(f)$ with white noise of variance σ_u^2. If the $b[k]$'s are zero, then we have the AR process introduced in Example 3.16. The estimation of the AR parameters of an AR model using an asymptotic MLE was discussed in Example 7.18. For the ARMA process even the asymptotic MLE proves to be intractable. An alternative approach relies on *equation error modeling* of the ACF. It estimates the AR filter parameters, leaving the MA filter parameters and white noise variance to be found by other means.

To determine the ACF we can take the inverse z transform of the PSD extended to the z plane or

$$\mathcal{P}_{xx}(z) = \frac{\sigma^2 \mathcal{B}(z)\mathcal{B}(z^{-1})}{\mathcal{A}(z)\mathcal{A}(z^{-1})}$$

where $B(f) = \mathcal{B}(\exp(j2\pi f))$, $A(f) = \mathcal{A}(\exp(j2\pi f))$. The usual PSD is obtained by evaluating $\mathcal{P}_{xx}(z)$ on the unit circle of the z plane or for $z = \exp(j2\pi f)$. It is convenient to develop a difference equation for the ACF. This difference equation will serve as the basis for the equation error modeling approach. Taking the inverse z transform of $\mathcal{A}(z)\mathcal{P}_{xx}(z)$, we have

$$\mathcal{Z}^{-1}\{\mathcal{A}(z)\mathcal{P}_{xx}(z)\} = \mathcal{Z}^{-1}\left\{\sigma^2 \mathcal{B}(z)\frac{\mathcal{B}(z^{-1})}{\mathcal{A}(z^{-1})}\right\}.$$

Since the filter impulse response is causal, it follows that

$$\begin{aligned} h[n] &= \mathcal{Z}^{-1}\left\{\frac{\mathcal{B}(z)}{\mathcal{A}(z)}\right\} \\ &= 0 \end{aligned}$$

for $n < 0$. Consequently, we have that

$$h[-n] = \mathcal{Z}^{-1}\left\{\frac{\mathcal{B}(z^{-1})}{\mathcal{A}(z^{-1})}\right\} = 0$$

for $n > 0$ and hence is anticausal. Thus,

$$\begin{aligned} \mathcal{Z}^{-1}\left\{\sigma^2\mathcal{B}(z)\frac{\mathcal{B}(z^{-1})}{\mathcal{A}(z^{-1})}\right\} &= \sigma^2 b[n] \star h[-n] \\ &= 0 \qquad \text{for } n > q. \end{aligned}$$

Continuing, we have

$$\begin{aligned} \mathcal{Z}^{-1}\left\{\mathcal{A}(z)\mathcal{P}_{xx}(z)\right\} &= \mathcal{Z}^{-1}\left\{\sigma^2\mathcal{B}(z)\frac{\mathcal{B}(z^{-1})}{\mathcal{A}(z^{-1})}\right\} \\ &= 0 \qquad \text{for } n > q. \end{aligned}$$

Finally, a difference equation for the ACF can be written as

$$\sum_{k=0}^{p} a[k] r_{xx}[n-k] = 0 \qquad \text{for } n > q \tag{8.63}$$

where $a[0] = 1$. These equations are called the *modified Yule-Walker equations*. Recall from Example 7.18 that the Yule-Walker equations for an AR process were identical except that they held for $n > 0$ (since $q = 0$ for an AR process). Only the AR filter parameters appear in these equations.

In practice, we must estimate the ACF lags as

$$\hat{r}_{xx}[k] = \frac{1}{N}\sum_{n=0}^{N-1-|k|} x[n]x[n+|k|],$$

assuming $x[n]$ is available for $n = 0, 1, \ldots, N-1$. Substituting these into (8.63) yields

$$\sum_{k=0}^{p} a[k]\hat{r}_{xx}[n-k] = \epsilon[n] \qquad n > q$$

where $\epsilon[n]$ denotes the error due to the effect of errors in the ACF function estimate. The model becomes

$$\hat{r}_{xx}[n] = -\sum_{k=1}^{p} a[k]\hat{r}_{xx}[n-k] + \epsilon[n] \qquad n > q$$

which can be seen to be linear in the unknown AR filter parameters. If the ACF is estimated for lags $n = 0, 1, \ldots, M$ (where we must have $M \leq N-1$), then a LSE of $a[k]$ will minimize

$$\begin{aligned} J &= \sum_{n=q+1}^{M}\left[\hat{r}_{xx}[n] - \left(-\sum_{k=1}^{p} a[k]\hat{r}_{xx}[n-k]\right)\right]^2 \\ &= (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T(\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) \end{aligned} \tag{8.64}$$

where

$$\mathbf{x} = \begin{bmatrix} \hat{r}_{xx}[q+1] \\ \hat{r}_{xx}[q+2] \\ \vdots \\ \hat{r}_{xx}[M] \end{bmatrix}$$

$$\boldsymbol{\theta} = \begin{bmatrix} a[1] \\ a[2] \\ \vdots \\ a[p] \end{bmatrix}$$

$$\mathbf{H} = -\begin{bmatrix} \hat{r}_{xx}[q] & \hat{r}_{xx}[q-1] & \dots & \hat{r}_{xx}[q-p+1] \\ \hat{r}_{xx}[q+1] & \hat{r}_{xx}[q] & \dots & \hat{r}_{xx}[q-p+2] \\ \vdots & \vdots & \vdots & \vdots \\ \hat{r}_{xx}[M-1] & \hat{r}_{xx}[M-2] & \dots & \hat{r}_{xx}[M-p] \end{bmatrix}.$$

The LSE of $\boldsymbol{\theta}$ is $(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}$ and is termed the *least squares modified Yule-Walker equations*. It is interesting to observe that in this problem we use a LSE for the estimated ACF data, not the original data. Additionally, the observation matrix $\mathbf{H}$, which is usually a known deterministic matrix, is now a random matrix. As expected, the statistics of the LSE are difficult to determine. In practice, M should not be chosen too large since the ACF estimate is less reliable at higher lags due to the $(N-k)$ lag products averaged in the estimate of $r_{xx}[k]$. Some researchers advocate a weighted LSE to reflect the tendency for the errors of (8.64) to increase as n increases. As an example, the choice $w_n = 1 - (n/(M+1))$ might be made and the weighted LSE of (8.16) implemented. Further discussions of this problem can be found in [Kay 1988]. ◇

Example 8.13 - Adaptive Noise Canceler

A common problem in signal processing is to reduce unwanted noise. It arises in such diverse problems as the suppression of a 60 Hz interference in an electronic circuit and the suppression of a mother's heartbeat masking the fetal heartbeat in an EKG trace [Widrow and Stearns 1985]. A particularly effective means of accomplishing this is the adaptive noise canceler (ANC). It assumes the availability of a reference noise that after suitable filtering can be subtracted from the noise that is to be canceled. As an example, to cancel a 60 Hz interference we need to know the amplitude and phase of the interference, both of which are generally unknown. A suitable adaptive filter as shown in Figure 8.14 can modify the amplitude and phase so as to make it nearly identical to the interference to be canceled. A discrete-time ANC is shown in Figure 8.15. The primary channel contains noise $x[n]$ to be canceled. To effect cancelation a reference channel containing a known sequence $x_R[n]$ that is similar but not identical to the noise is then filtered to improve the match. The filter $\mathcal{H}_n(z)$ has weights that are determined

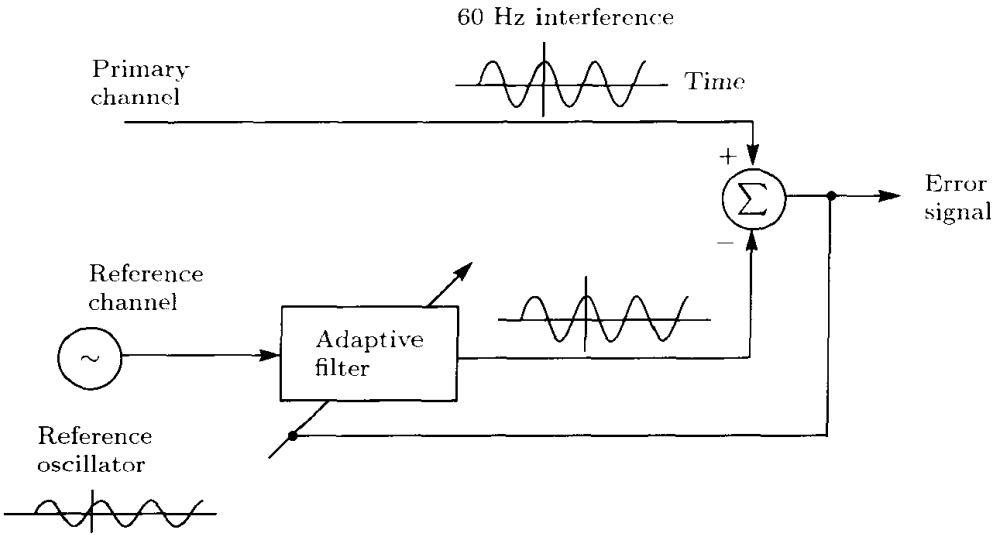


Figure 8.14 Adaptive noise canceler for 60 Hz interference

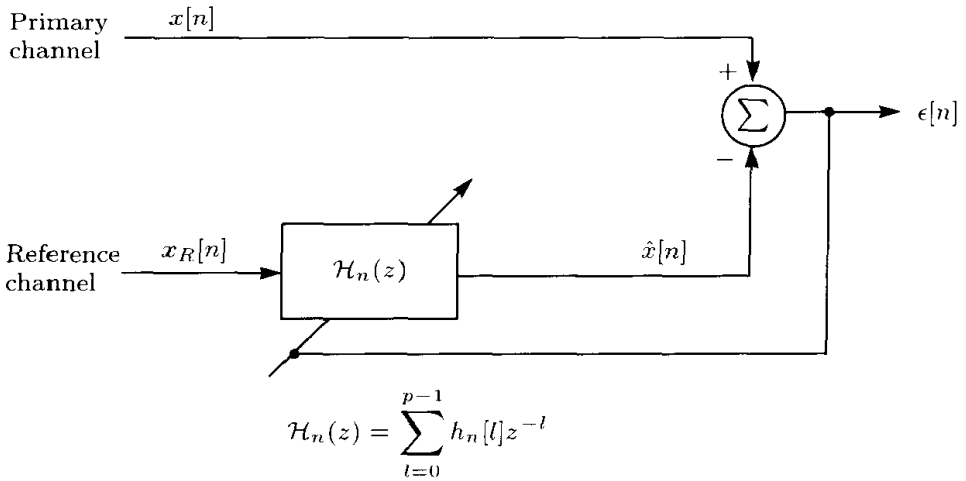
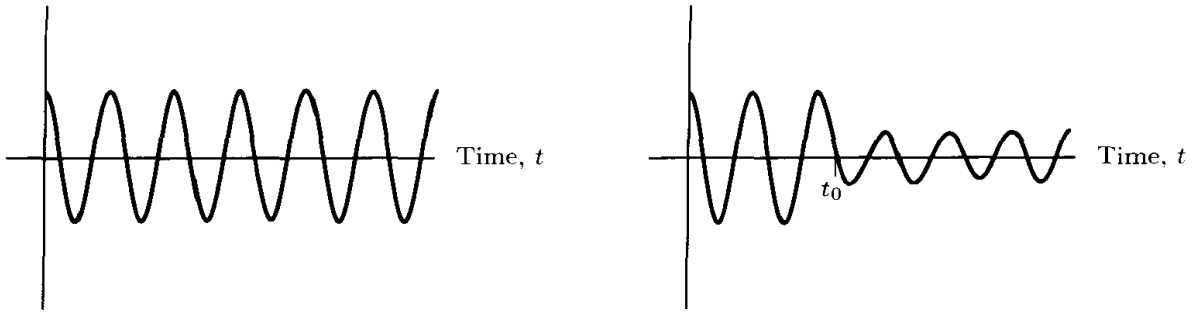


Figure 8.15 Generic adaptive noise canceler

at each time n to make $\hat{x}[k] \approx x[k]$ for $k = 0, 1, \ldots, n$. Since we want $\epsilon[n] \approx 0$, it makes sense to choose the weights at time n to minimize

$$\begin{aligned} J[n] &= \sum_{k=0}^{n} \epsilon^2[k] \\ &= \sum_{k=0}^{n} \left(x[k] - \hat{x}[k]\right)^2 \\ &= \sum_{k=0}^{n} \left(x[k] - \sum_{l=0}^{p-1} h_n[l] x_R[k-l] \right)^2 \end{aligned}$$



(a) Fixed interference model (stationary)

(b) Changing interference model (nonstationary)

Figure 8.16 Stationarity of interference model

where $h_n[l]$ are the filter weights at time n. It is immediately recognized that the weights can be determined as the solution of a sequential LS problem. Before giving the solution we remark that the LSE assumes that the interference is fixed over time or that it is stationary. For instance, the primary channel containing the sinusoid is assumed to appear as shown in Figure 8.16a. If, however, the interference appears as in Figure 8.16b, then the adaptive filter must quickly change its coefficients to respond to the changing interference for $t > t_0$. How fast it can change depends upon how many error terms are included in $J[n]$ before and after the transition at $t = t_0$. The chosen weights for $t > t_0$ will be a compromise between the old weights and the desired new ones. If the transition occurs in discrete time at $n = n_0$, then we should probably expect the wrong weights until $n \gg n_0$. To allow the filter to adapt more quickly we can downweight previous errors in $J[n]$ by incorporating a weighting or "forgetting factor" λ, where $0 < \lambda < 1$, as follows:

$$J[n] = \sum_{k=0}^{n} \lambda^{n-k} \left(x[k] - \sum_{l=0}^{p-1} h_n[l] x_R[k-l] \right)^2 .$$

This modification downweights previous errors exponentially, allowing the filter to react more quickly to interference changes. The penalty paid is that the estimates of the filter weights are noisier, relying on fewer effective LS errors. Note that the solution will not change if we minimize instead

$$J'[n] = \sum_{k=0}^{n} \frac{1}{\lambda^k} \left(x[k] - \sum_{l=0}^{p-1} h_n[l] x_R[k-l] \right)^2 \tag{8.65}$$

for each n. Now we have our standard sequential *weighted* LS problem described in Section 8.7. Referring to (8.46), we identify the sequential LSE of the filter weights as

$$\hat{\boldsymbol{\theta}}[n] = \left[\begin{array}{cccc} \hat{h}_n[0] & \hat{h}_n[1] & \ldots & \hat{h}_n[p-1] \end{array} \right]^T .$$

The data vector $\mathbf{h}[n]$ is given as (not to be confused with the impulse response $h[n]$)

$$\mathbf{h}[n] = \begin{bmatrix} x_R[n] & x_R[n-1] & \dots & x_R[n-p+1] \end{bmatrix}^T$$

and the weights σ_n^2 are given by λ^n. Also, note that

$$\begin{aligned} e[n] &= x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1] \\ &= x[n] - \sum_{l=0}^{p-1} \hat{h}_{n-1}[l] x_R[n-l] \end{aligned}$$

is the error at time n based on the filter weights at time $n-1$. (This is not the same as $\epsilon[n] = x[n] - \hat{x}[n] = x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n]$.) The algorithm is summarized from (8.46)–(8.48) as

$$\hat{\boldsymbol{\theta}}[n] = \hat{\boldsymbol{\theta}}[n-1] + \mathbf{K}[n]e[n]$$

where

$$\begin{aligned} e[n] &= x[n] - \sum_{l=0}^{p-1} \hat{h}_{n-1}[l] x_R[n-l] \\ \mathbf{K}[n] &= \frac{\boldsymbol{\Sigma}[n-1]\mathbf{h}[n]}{\lambda^n + \mathbf{h}^T[n]\boldsymbol{\Sigma}[n-1]\mathbf{h}[n]} \\ \mathbf{h}[n] &= \begin{bmatrix} x_R[n] & x_R[n-1] & \dots & x_R[n-p+1] \end{bmatrix}^T \\ \boldsymbol{\Sigma}[n] &= \left(\mathbf{I} - \mathbf{K}[n]\mathbf{h}^T[n]\right)\boldsymbol{\Sigma}[n-1]. \end{aligned}$$

The forgetting factor is chosen as $0 < \lambda < 1$, with λ near 1 or $0.9 < \lambda < 1$ being typical. As an example, for a sinusoidal interference

$$x[n] = 10\cos(2\pi(0.1)n + \pi/4)$$

and a reference signal

$$x_R[n] = \cos(2\pi(0.1)n)$$

we implement the ANC using two filter coefficients ($p = 2$) since the reference signal need only be modified in amplitude and phase to match the interference. To initialize the sequential LS estimator we choose

$$\begin{aligned} \hat{\boldsymbol{\theta}}[-1] &= \mathbf{0} \\ \boldsymbol{\Sigma}[-1] &= 10^5\mathbf{I} \end{aligned}$$

and a forgetting factor of $\lambda = 0.99$ is assumed. The interference $x[n]$ and output $\epsilon[n]$ of the ANC are shown in Figures 8.17a and 8.17b, respectively. As expected, the interference is canceled. The LSE of the filter coefficients is shown in Figure 8.17c and is seen to rapidly converge to the steady-state value. To find the steady state values of the weights we need to solve

$$\mathcal{H}[\exp(2\pi(0.1))] = 10\exp(j\pi/4)$$

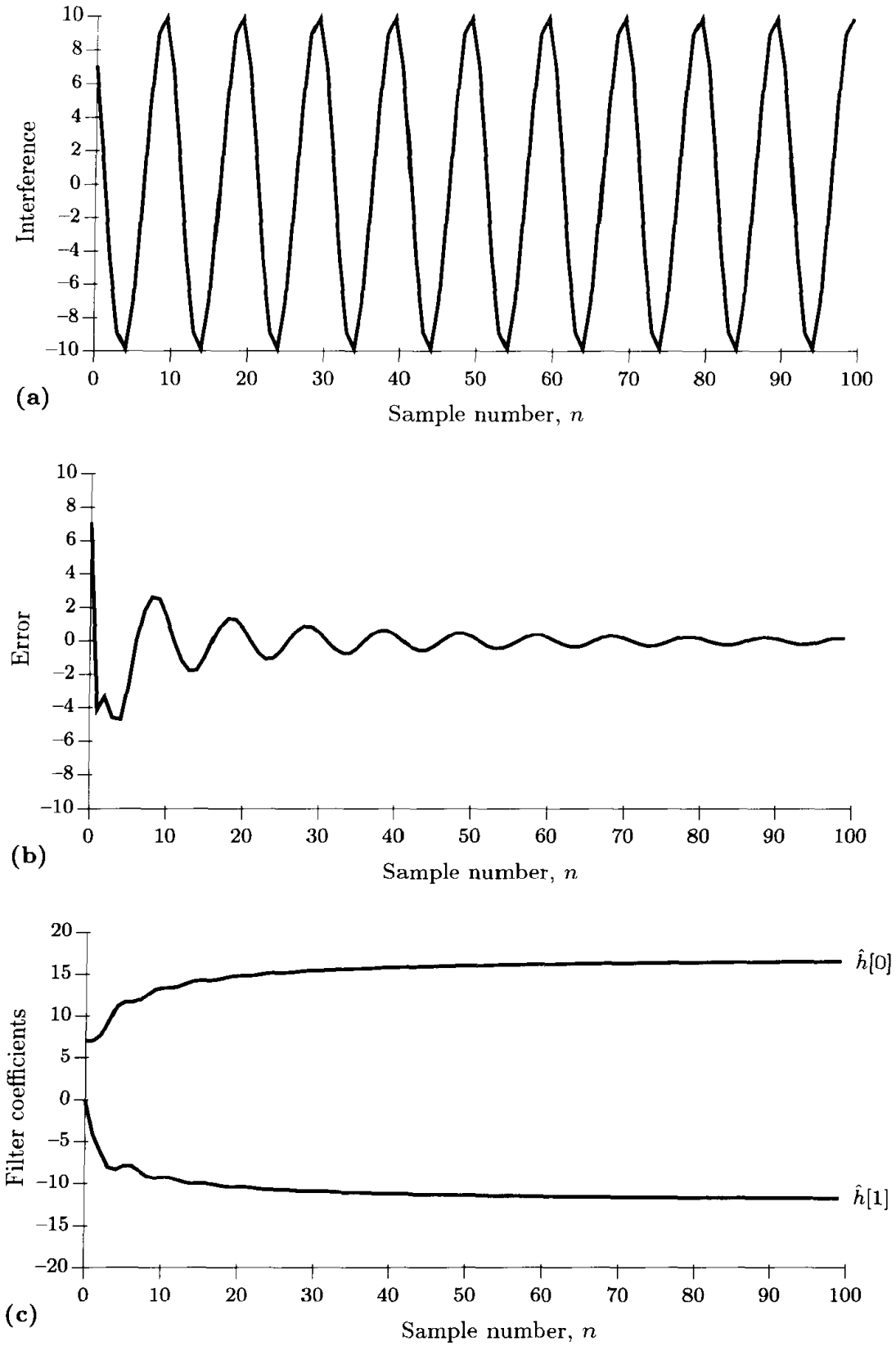


Figure 8.17 Interference cancellation example

which says that the adaptive filter must increase the gain of the reference signal by 10 and the phase by $\pi/4$ to match the interference. Solving, we have

$$h[0] + h[1]\exp(-j2\pi(0.1)) = 10\exp(j\pi/4)$$

which results in $h[0] = 16.8$ and $h[1] = -12.0$. ◇

Example 8.14 - Phase-Locked Loop

We now consider the problem of carrier recovery in a communication system, which is necessary for coherent demodulation. It is assumed that the carrier is received but is embedded in noise. The received noise-free carrier is

$$s[n] = \cos(2\pi f_0 n + \phi) \qquad n = -M, \ldots, 0, \ldots, M$$

where the frequency f_0 and phase ϕ are to be estimated. A symmetric observation interval is chosen to simplify the algebra. A LSE is to be used. Due to the nonlinearity we employ the linearization technique of (8.62). Determining first $\mathbf{H}(\boldsymbol{\theta})$, where $\boldsymbol{\theta} = [f_0\ \phi]^T$, we have

$$\begin{aligned} \frac{\partial s[n]}{\partial f_0} &= -n2\pi \sin(2\pi f_0 n + \phi) \\ \frac{\partial s[n]}{\partial \phi} &= -\sin(2\pi f_0 n + \phi) \end{aligned}$$

so that

$$\mathbf{H}(\boldsymbol{\theta}) = -\begin{bmatrix} -M2\pi \sin(-2\pi f_0 M + \phi) & \sin(-2\pi f_0 M + \phi) \\ -(M-1)2\pi \sin(-2\pi f_0 (M-1) + \phi) & \sin(-2\pi f_0 (M-1) + \phi) \\ \vdots & \vdots \\ M2\pi \sin(2\pi f_0 M + \phi) & \sin(2\pi f_0 M + \phi) \end{bmatrix}$$

and

$$\mathbf{H}^T(\boldsymbol{\theta})\mathbf{H}(\boldsymbol{\theta}) = \begin{bmatrix} 4\pi^2 \sum_{n=-M}^{M} n^2 \sin^2(2\pi f_0 n + \phi) & 2\pi \sum_{n=-M}^{M} n \sin^2(2\pi f_0 n + \phi) \\ 2\pi \sum_{n=-M}^{M} n \sin^2(2\pi f_0 n + \phi) & \sum_{n=-M}^{M} \sin^2(2\pi f_0 n + \phi) \end{bmatrix}.$$

But

$$\sum_{n=-M}^{M} n^2 \sin^2(2\pi f_0 n + \phi) = \sum_{n=-M}^{M} \left[\frac{n^2}{2} - \frac{n^2}{2}\cos(4\pi f_0 n + 2\phi)\right]$$

$$\sum_{n=-M}^{M} n\sin^2(2\pi f_0 n+\phi) = \sum_{n=-M}^{M}\left[\frac{n}{2}-\frac{n}{2}\cos(4\pi f_0 n+2\phi)\right]$$
$$\sum_{n=-M}^{M} \sin^2(2\pi f_0 n+\phi) = \sum_{n=-M}^{M}\left[\frac{1}{2}-\frac{1}{2}\cos(4\pi f_0 n+2\phi)\right]$$

and since [Stoica et al. 1989]

$$\frac{1}{(2M+1)^{i+1}}\sum_{n=-M}^{M} n^i\cos(4\pi f_0 n+2\phi)\approx 0 \qquad i=0,1,2$$

we have from (3.22)

$$\mathbf{H}^T(\boldsymbol{\theta})\mathbf{H}(\boldsymbol{\theta}) \approx \begin{bmatrix} \frac{4\pi^2}{3}M(M+1)(2M+1) & 0 \\ 0 & \frac{2M+1}{2} \end{bmatrix}$$
$$\approx \begin{bmatrix} \frac{8\pi^2M^3}{3} & 0 \\ 0 & M \end{bmatrix}$$

for $M \gg 1$. We have then from (8.62)

$$f_{0_{k+1}} = f_{0_k} - \frac{3}{4\pi M^3}\sum_{n=-M}^{M} n\sin(2\pi f_{0_k}n+\phi_k)\left(x[n]-\cos(2\pi f_{0_k}n+\phi_k)\right)$$
$$\phi_{k+1} = \phi_k - \frac{1}{M}\sum_{n=-M}^{M}\sin(2\pi f_{0_k}n+\phi_k)\left(x[n]-\cos(2\pi f_{0_k}n+\phi_k)\right)$$

or since

$$\frac{1}{2M+1}\sum_{n=-M}^{M}\sin(2\pi f_{0_k}n+\phi_k)\cos(2\pi f_{0_k}n+\phi_k)$$
$$= \frac{1}{2(2M+1)}\sum_{n=-M}^{M}\sin(4\pi f_{0_k}n+2\phi_k)$$
$$\approx 0$$

it follows that

$$f_{0_{k+1}} = f_{0_k} - \frac{3}{4\pi M^3}\sum_{n=-M}^{M} nx[n]\sin(2\pi f_{0_k}n+\phi_k)$$
$$\phi_{k+1} = \phi_k - \frac{1}{M}\sum_{n=-M}^{M} x[n]\sin(2\pi f_{0_k}n+\phi_k).$$

The LSE of frequency and phase can be implemented as a phase-locked loop [Proakis 1983]. In Problem 8.29 it is shown that at convergence for a high enough SNR the solution will be

$$\begin{aligned} f_{0_{k+1}} &= f_{0_k} \approx f_0 \\ \phi_{k+1} &= \phi_k \approx \phi. \end{aligned}$$

◇

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Problems

8.1 The LS error

$$J = \sum_{n=0}^{N-1} (x[n] - A\cos 2\pi f_0 n - Br^n)^2$$

is to be minimized to find the LSE of $\boldsymbol{\theta} = [A\, f_0\, B\, r]^T$, where $0 < r < 1$. Is this a linear or nonlinear LS problem? Is the LS error quadratic in any of the parameters, and if so, which ones? How could you solve this minimization problem using a digital computer?

8.2 Show that the inequality given in (8.7) holds.

8.3 For the signal model

$$s[n] = \begin{cases} A & 0 \le n \le M-1 \\ -A & M \le n \le N-1, \end{cases}$$

find the LSE of A and the minimum LS error. Assume that $x[n] = s[n] + w[n]$ for $n = 0, 1, \ldots, N-1$ are observed. If now $w[n]$ is WGN with variance σ^2, find the PDF of the LSE.

8.4 Derive the LSE for a vector parameter as given by (8.10) by verifying the identity

$$J = (\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}})^T(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}) + (\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})^T\mathbf{H}^T\mathbf{H}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})$$

where

$$\hat{\boldsymbol{\theta}} = (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{x}.$$

To complete the proof show that J is minimized when $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$, assuming that $\mathbf{H}$ is full rank and therefore that $\mathbf{H}^T\mathbf{H}$ is positive definite.

8.5 For the signal model

$$s[n] = \sum_{i=1}^{p} A_i \cos 2\pi f_i n$$

where the frequencies f_i are known and the amplitudes A_i are to be estimated, find the LSE *normal equations* (do not attempt to solve them). Then, if the frequencies are specifically known to be $f_i = i/N$, explicitly find the LSE and the minimum LS error. Finally, if $x[n] = s[n] + w[n]$, where $w[n]$ is WGN with variance σ^2, determine the PDF of the LSE, assuming the given frequencies. Hint: The columns of $\mathbf{H}$ are orthogonal for the given frequencies.

8.6 For the general LSE of (8.10) find the PDF of the LSE if it is known that $\mathbf{x} \sim \mathcal{N}(\mathbf{H}\boldsymbol{\theta}, \sigma^2\mathbf{I})$. Is the LSE unbiased?

8.7 In this problem we consider the estimation of the noise variance σ^2 in the model $\mathbf{x} = \mathbf{H}\boldsymbol{\theta} + \mathbf{w}$, where $\mathbf{w}$ is zero mean with covariance matrix $\sigma^2\mathbf{I}$. The estimator

$$\hat{\sigma^2} = \frac{1}{N}J_{\min} = \frac{1}{N}(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}})^T(\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}})$$

where $\hat{\boldsymbol{\theta}}$ is the LSE of (8.10), is proposed. Is this estimator unbiased, and if not, can you propose one that is? Explain your results. Hint: The identities $E(\mathbf{x}^T\mathbf{y}) = E(\text{tr}(\mathbf{y}\mathbf{x}^T)) = \text{tr}(E(\mathbf{y}\mathbf{x}^T))$ and $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ will be useful.

8.8 Verify the LSE for A given in (8.15). Find the mean and variance for $\hat{A}$ if $x[n] = A + w[n]$, where $w[n]$ is zero mean uncorrelated noise with variance σ_n^2.

8.9 Verify (8.16) and (8.17) for the weighted LSE by noting that if $\mathbf{W}$ is positive definite, we can write it as $\mathbf{W} = \mathbf{D}^T\mathbf{D}$, where $\mathbf{D}$ is an invertible $N \times N$ matrix.

8.10 Referring to Figure 8.2b, prove that

$$||\hat{\mathbf{s}}||^2 + ||\mathbf{x} - \hat{\mathbf{s}}||^2 = ||\mathbf{x}||^2.$$

This can be thought of as the least squares Pythagorean theorem.

8.11 In this problem we prove that a projection matrix $\mathbf{P}$ must be symmetric. Let $\mathbf{x} = \boldsymbol{\xi} + \boldsymbol{\xi}^{\perp}$, where $\boldsymbol{\xi}$ lies in a subspace which is the range of the projection matrix or $\mathbf{P}\mathbf{x} = \boldsymbol{\xi}$ and $\boldsymbol{\xi}^{\perp}$ lies in the orthogonal subspace or $\mathbf{P}\boldsymbol{\xi}^{\perp} = \mathbf{0}$. For arbitrary vectors $\mathbf{x}_1, \mathbf{x}_2$ in R^N show that

$$\mathbf{x}_1^T \mathbf{P} \mathbf{x}_2 - \mathbf{x}_2^T \mathbf{P} \mathbf{x}_1 = 0$$

by decomposing $\mathbf{x}_1$ and $\mathbf{x}_2$ as discussed above. Finally, prove the desired result.

8.12 Prove the following properties of the projection matrix

$$\mathbf{P} = \mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T.$$

a. $\mathbf{P}$ is idempotent.

b. $\mathbf{P}$ is positive semidefinite.

c. The eigenvalues of $\mathbf{P}$ are either 1 or 0.

d. The rank of $\mathbf{P}$ is p. Use the fact that the trace of a matrix is equal to the sum of its eigenvalues.

8.13 Verify the result given in (8.24).

8.14 In this problem we derive the order-update LS equations using a geometrical argument. Assume that $\hat{\boldsymbol{\theta}}_k$ is available and therefore

$$\hat{\mathbf{s}}_k = \mathbf{H}_k \hat{\boldsymbol{\theta}}_k$$

is known. Now, if $\mathbf{h}_{k+1}$ is used, the LS signal estimate becomes

$$\hat{\mathbf{s}}_{k+1} = \hat{\mathbf{s}}_k + \alpha \mathbf{h}'_{k+1}$$

where $\mathbf{h}'_{k+1}$ is the component of $\mathbf{h}_{k+1}$ orthogonal to the subspace spanned by $\{\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_k\}$. First, find $\mathbf{h}'_{k+1}$ and then determine α by noting that $\alpha\mathbf{h}'_{k+1}$ is the projection of $\mathbf{x}$ onto $\mathbf{h}'_{k+1}$. Finally, since

$$\hat{\mathbf{s}}_{k+1} = \mathbf{H}_{k+1} \hat{\boldsymbol{\theta}}_{k+1}$$

determine $\hat{\boldsymbol{\theta}}_{k+1}$.

8.15 Use order-recursive LS to find the values of A and B that minimize

$$J = \sum_{n=0}^{N-1} (x[n] - A - Br^n)^2.$$

The parameter r is assumed to be known.

8.16 Consider the models

$$\begin{aligned} s_1[n] &= A \quad 0 \le n \le N-1 \\ s_2[n] &= \begin{cases} A & 0 \le n \le M-1 \\ B & M \le n \le N-1. \end{cases} \end{aligned}$$

The second signal is useful for modeling a jump in level at $n = M$. We can express the second model in the alternative form

$$s[n] = Au_1[n] + (B - A)u_2[n]$$

where

$$\begin{aligned} u_1[n] &= 1 \quad 0 \le n \le N-1 \\ u_2[n] &= \begin{cases} 0 & 0 \le n \le M-1 \\ 1 & M \le n \le N-1. \end{cases} \end{aligned}$$

Now let $\boldsymbol{\theta}_1 = A$ and $\boldsymbol{\theta}_2 = [A \ (B - A)]^T$ and find the LSE and minimum LS error for each model using order-recursive LS. Discuss how you might use it to detect a jump in level.

8.17 Prove that $\mathbf{P}_k^{\perp}\mathbf{x}$ is orthogonal to the space spanned by the columns of $\mathbf{H}_k$.

8.18 Using the orthogonal recursive projection matrix formula (8.34), derive the update formula for the minimum LS error (8.31).

8.19 Verify the sequential formula for the minimum LS error (8.43) by using the sequential LSE update (8.40).

8.20 Let $x[n] = Ar^n + w[n]$, where $w[n]$ is WGN with variance $\sigma^2 = 1$. Find the sequential LSE for A, assuming that r is known. Also, determine the variance of the LSE in sequential form and then solve explicitly for the variance as a function of n. Let $\hat{A}[0] = x[0]$ and $\text{var}(\hat{A}[0]) = \sigma^2 = 1$.

8.21 Using the sequential update formulas for the gain (8.41) and variance (8.42), solve for the gain and variance sequences if $\sigma_n^2 = r^n$. Use $\text{var}(\hat{A}[0]) = \text{var}(x[0]) = \sigma_0^2 = 1$ for initialization. Then, examine what happens to the gain and variance as $N \to \infty$ if $r = 1$, $0 < r < 1$, and $r > 1$. Hint: Solve for $1/\text{var}(\hat{A}[N])$.

8.22 By implementing a Monte Carlo computer simulation plot $\hat{A}[N]$ as given by (8.40). Assume that the data are given by

$$x[n] = A + w[n]$$

where $w[n]$ is zero mean WGN with $\sigma_n^2 = r^n$. Use $A = 10$ and $r = 1, 0.95, 1.05$. Initialize the estimator by using $\hat{A}[0] = x[0]$ and $\text{var}(\hat{A}[0]) = \text{var}(x[0]) = \sigma_0^2 = 1$. Also, plot the gain and variance sequences.

8.23 In this problem we examine the initialization of a sequential LSE. Assume that we choose $\hat{\boldsymbol{\theta}}[-1]$ and $\boldsymbol{\Sigma}[-1] = \alpha\mathbf{I}$ to initialize the LSE. We will show that, as $\alpha \to \infty$, the batch LSE

$$\begin{aligned}\hat{\boldsymbol{\theta}}_B[n] &= \left(\mathbf{H}^T[n]\mathbf{C}^{-1}[n]\mathbf{H}[n]\right)^{-1}\mathbf{H}^T[n]\mathbf{C}^{-1}[n]\mathbf{x}[n] \\ &= \left(\sum_{k=0}^{n}\frac{1}{\sigma_k^2}\mathbf{h}[k]\mathbf{h}^T[k]\right)^{-1}\left(\sum_{k=0}^{n}\frac{1}{\sigma_k^2}x[k]\mathbf{h}^T[k]\right)\end{aligned}$$

for $n \geq p$ is identical to the sequential LSE. First, assume that the initial observation vectors $\{\mathbf{h}[-p], \mathbf{h}[-(p-1)], \ldots, \mathbf{h}[-1]\}$ and noise variances $\{\sigma_{-p}^2, \sigma_{-(p-1)}^2, \ldots, \sigma_{-1}^2\}$ exist, so that for the chosen initial LSE and covariance we can use a batch estimator. Hence,

$$\hat{\boldsymbol{\theta}}[-1] = \hat{\boldsymbol{\theta}}_B[-1] = \left(\mathbf{H}^T[-1]\mathbf{C}^{-1}[-1]\mathbf{H}[-1]\right)^{-1}\mathbf{H}^T[-1]\mathbf{C}^{-1}[-1]\mathbf{x}[-1]$$

and

$$\boldsymbol{\Sigma}[-1] = \left(\mathbf{H}^T[-1]\mathbf{C}^{-1}[-1]\mathbf{H}[-1]\right)^{-1}$$

where

$$\begin{aligned}\mathbf{H}[-1] &= \begin{bmatrix}\mathbf{h}^T[-p] \\ \mathbf{h}^T[-(p-1)] \\ \vdots \\ \mathbf{h}^T[-1]\end{bmatrix} \\ \mathbf{C}[-1] &= \text{diag}\left(\sigma_{-p}^2, \sigma_{-(p-1)}^2, \ldots, \sigma_{-1}^2\right).\end{aligned}$$

Thus, we may view the initial estimate for the sequential LSE as the result of applying a batch estimator to the initial observation vectors. Since the sequential LSE initialized using a batch estimator is identical to the batch LSE using all the observation vectors, we have for the sequential LSE with the assumed initial conditions

$$\hat{\boldsymbol{\theta}}_S[n] = \left(\sum_{k=-p}^{n}\frac{1}{\sigma_k^2}\mathbf{h}[k]\mathbf{h}^T[k]\right)^{-1}\left(\sum_{k=-p}^{n}\frac{1}{\sigma_k^2}x[k]\mathbf{h}^T[k]\right).$$

Prove that this can be rewritten as

$$\hat{\boldsymbol{\theta}}_S[n] =$$

$$\left(\boldsymbol{\Sigma}^{-1}[-1] + \sum_{k=0}^{n}\frac{1}{\sigma_k^2}\mathbf{h}[k]\mathbf{h}^T[k]\right)^{-1}\left(\boldsymbol{\Sigma}^{-1}[-1]\hat{\boldsymbol{\theta}}[-1] + \sum_{k=0}^{n}\frac{1}{\sigma_k^2}x[k]\mathbf{h}^T[k]\right).$$

Then examine what happens as $\alpha \to \infty$ for $0 \leq n \leq p-1$ and $n \geq p$.

8.24 In Example 8.8 determine $\hat{\mathbf{s}}_c$ by first projecting $\mathbf{x}$ onto the subspace spanned by $\mathbf{h}_1$ and $\mathbf{h}_2$ to produce $\hat{\mathbf{s}}$, and then projecting $\hat{\mathbf{s}}$ onto the constrained subspace.

8.25 If the signal model is

$$s[n] = A + B(-1)^n \qquad n = 0, 1, \ldots, N-1$$

and N is even, find the LSE of $\boldsymbol{\theta} = [A\, B]^T$. Now assume that $A = B$ and repeat the problem using a constrained LS approach. Compare your results.

8.26 Consider the minimization of $h(\theta)$ with respect to θ and assume that

$$\hat{\theta} = \arg\min_{\theta} h(\theta).$$

Let $\alpha = g(\theta)$, where g is a one-to-one mapping. Prove that if $\hat{\alpha}$ minimizes $h(g^{-1}(\alpha))$, then $\hat{\theta} = g^{-1}(\hat{\alpha})$.

8.27 Let the observed data be

$$x[n] = \exp(\theta) + w[n] \qquad n = 0, 1, \ldots, N-1.$$

Set up a Newton-Raphson iteration to find the LSE of θ. Can you avoid the nonlinear optimization and find the LSE analytically?

8.28 In Example 8.11 the true LSE must minimize

$$J = \sum_{n=0}^{N-1} (h_d[n] - h[n])^2.$$

In this problem we derive the equations that must be optimized to find the true LSE [Scharf 1991]. It is assumed that $p = q+1$. First, show that the signal model can be written as

$$\mathbf{s} = \begin{bmatrix} h[0] \\ h[1] \\ \vdots \\ h[N-1] \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} g[0] & 0 & \cdots & 0 \\ g[1] & g[0] & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ g[q] & g[q-1] & \cdots & g[0] \\ g[q+1] & g[q] & \cdots & g[1] \\ \vdots & \vdots & \vdots & \vdots \\ g[N-1] & g[N-2] & \cdots & g[N-1-q] \end{bmatrix}}_{\mathbf{G}} \underbrace{\begin{bmatrix} b[0] \\ b[1] \\ \vdots \\ b[q] \end{bmatrix}}_{\mathbf{b}}$$

where

$$g[n] = \mathcal{Z}^{-1}\left\{\frac{1}{A(z)}\right\}$$

and $\mathbf{G}$ is $N \times (q+1)$. Letting $\mathbf{x} = [h_d[0]\, h_d[1] \ldots h_d[N-1]]^T$, show that

$$J(\mathbf{a}, \hat{\mathbf{b}}) = \mathbf{x}^T \left(\mathbf{I} - \mathbf{G}(\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T\right)\mathbf{x}$$

where $\hat{\mathbf{b}}$ is the LSE of $\mathbf{b}$ for a given $\mathbf{a}$. Finally, prove that

$$\mathbf{I} - \mathbf{G}(\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T = \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \tag{8.66}$$

where $\mathbf{A}^T$ is the $(N-q-1) \times N = (N-p) \times N$ matrix

$$\mathbf{A}^T = \begin{bmatrix} a[p] & a[p-1] & \ldots & 1 & 0 & 0 & \ldots & 0 \\ 0 & a[p] & \ldots & a[1] & 1 & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \ldots & 0 & a[p] & a[p-1] & \ldots & 1 \end{bmatrix}.$$

Hence, the LSE of $\mathbf{a}$ is found by minimizing

$$\mathbf{x}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{x}.$$

Once this is found (using a nonlinear optimization), we have

$$\begin{aligned} \hat{\mathbf{b}} &= (\mathbf{G}^T\mathbf{G})^{-1}\mathbf{G}^T\mathbf{x} \\ &= \left(\mathbf{I} - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\right)\mathbf{x} \end{aligned}$$

where the elements in $\mathbf{A}$ are replaced by the LSE of $\mathbf{a}$. Note that this problem is an example of a separable nonlinear LS problem. Hint: To prove (8.66) consider $\mathbf{L} = [\mathbf{A}\,\mathbf{G}]$, which is invertible (since it is full rank), and compute

$$\mathbf{L}(\mathbf{L}^T\mathbf{L})^{-1}\mathbf{L}^T = \mathbf{I}.$$

You will also need to observe that $\mathbf{A}^T\mathbf{G} = \mathbf{0}$, which follows from $a[n] \star g[n] = \delta[n]$.

8.29 In Example 8.14 assume that the phase-locked loop converges so that $f_{0_{k+1}} = f_{0_k}$ and $\phi_{k+1} = \phi_k$. For a high SNR so that $x[n] \approx \cos(2\pi f_0 n + \phi)$, show that the final iterates will be the true values of frequency and phase.

Appendix 8A

Derivation of Order-Recursive Least Squares

In this appendix we derive the order-recursive least squares formulas (8.28)–(8.31). Referring to (8.25), we have

$$\hat{\boldsymbol{\theta}}_{k+1} = (\mathbf{H}_{k+1}^T \mathbf{H}_{k+1})^{-1} \mathbf{H}_{k+1}^T \mathbf{x}.$$

But

$$\begin{aligned} \mathbf{H}_{k+1}^T \mathbf{H}_{k+1} &= \begin{bmatrix} \mathbf{H}_k^T \\ \mathbf{h}_{k+1}^T \end{bmatrix} \begin{bmatrix} \mathbf{H}_k & \mathbf{h}_{k+1} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{H}_k^T \mathbf{H}_k & \mathbf{H}_k^T \mathbf{h}_{k+1} \\ \mathbf{h}_{k+1}^T \mathbf{H}_k & \mathbf{h}_{k+1}^T \mathbf{h}_{k+1} \end{bmatrix} = \begin{bmatrix} k \times k & k \times 1 \\ 1 \times k & 1 \times 1 \end{bmatrix}. \end{aligned}$$

Using the inverse of a partitioned matrix formula (see Appendix 1)

$$\begin{bmatrix} \mathbf{A} & \mathbf{b} \\ \mathbf{b}^T & c \end{bmatrix}^{-1} = \begin{bmatrix} \left(\mathbf{A} - \dfrac{\mathbf{b}\mathbf{b}^T}{c}\right)^{-1} & -\dfrac{1}{c}\left(\mathbf{A} - \dfrac{\mathbf{b}\mathbf{b}^T}{c}\right)^{-1}\mathbf{b} \\ -\dfrac{1}{c}\mathbf{b}^T\left(\mathbf{A} - \dfrac{\mathbf{b}\mathbf{b}^T}{c}\right)^{-1} & \dfrac{1}{c - \mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}} \end{bmatrix}$$

for $\mathbf{A}$ a symmetric $k \times k$ matrix, $\mathbf{b}$ a $k \times 1$ vector, and c a scalar, and also Woodbury's identity (see Appendix 1)

$$\left(\mathbf{A} - \frac{\mathbf{b}\mathbf{b}^T}{c}\right)^{-1} = \mathbf{A}^{-1} + \frac{\mathbf{A}^{-1}\mathbf{b}\mathbf{b}^T\mathbf{A}^{-1}}{c - \mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}}$$

we have

$$(\mathbf{H}_{k+1}^T\mathbf{H}_{k+1})^{-1} = \begin{bmatrix} \underbrace{\mathbf{D}_k + \frac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k}{\mathbf{h}_{k+1}^T\mathbf{h}_{k+1} - \mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}}}_{\mathbf{E}_k} & -\frac{\mathbf{E}_k\mathbf{H}_k^T\mathbf{h}_{k+1}}{\mathbf{h}_{k+1}^T\mathbf{h}_{k+1}} \\ -\frac{\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{E}_k}{\mathbf{h}_{k+1}^T\mathbf{h}_{k+1}} & \frac{1}{\mathbf{h}_{k+1}^T\mathbf{h}_{k+1} - \mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}} \end{bmatrix}$$

where $\mathbf{D}_k = (\mathbf{H}_k^T\mathbf{H}_k)^{-1}$ or

$$\mathbf{D}_{k+1} = \begin{bmatrix} \mathbf{D}_k + \frac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} & -\frac{\mathbf{E}_k\mathbf{H}_k^T\mathbf{h}_{k+1}}{\mathbf{h}_{k+1}^T\mathbf{h}_{k+1}} \\ -\frac{\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{E}_k}{\mathbf{h}_{k+1}^T\mathbf{h}_{k+1}} & \frac{1}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \end{bmatrix}.$$

Simplifying the off-diagonal block of $\mathbf{D}_{k+1}$, we have

$$\begin{aligned} \frac{\mathbf{E}_k\mathbf{H}_k^T\mathbf{h}_{k+1}}{\mathbf{h}_{k+1}^T\mathbf{h}_{k+1}} &= \frac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}}{\mathbf{h}_{k+1}^T\mathbf{h}_{k+1}} + \frac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{h}_{k+1}} \\ &= \frac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\left[\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1} + \mathbf{h}_{k+1}^T(\mathbf{I}-\mathbf{P}_k^\perp)\mathbf{h}_{k+1}\right]}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{h}_{k+1}} \\ &= \frac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}}. \end{aligned}$$

Finally, we have

$$\begin{aligned} \mathbf{D}_{k+1} &= \begin{bmatrix} \mathbf{D}_k + \frac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} & -\frac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ -\frac{\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} & \frac{1}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \end{bmatrix} \\ &= \begin{bmatrix} k\times k & k\times 1 \\ 1\times k & 1\times 1 \end{bmatrix}. \end{aligned}$$

To find $\hat{\boldsymbol{\theta}}_{k+1}$ we make use of the previous result

$$\begin{aligned} \hat{\boldsymbol{\theta}}_{k+1} &= \mathbf{D}_{k+1}\mathbf{H}_{k+1}^T\mathbf{x} \\ &= \mathbf{D}_{k+1}\begin{bmatrix} \mathbf{H}_k^T\mathbf{x} \\ \mathbf{h}_{k+1}^T\mathbf{x} \end{bmatrix} \\ &= \begin{bmatrix} \mathbf{D}_k\mathbf{H}_k^T\mathbf{x} + \frac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} - \frac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ -\frac{\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} + \frac{\mathbf{h}_{k+1}^T\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \end{bmatrix} \end{aligned}$$

$$= \begin{bmatrix} \hat{\boldsymbol{\theta}}_k - \dfrac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T(\mathbf{x}-\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T\mathbf{x})}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ \dfrac{\mathbf{h}_{k+1}^T(\mathbf{x}-\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T\mathbf{x})}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \end{bmatrix}$$

$$= \begin{bmatrix} \hat{\boldsymbol{\theta}}_k - \dfrac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ \dfrac{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \end{bmatrix}.$$

Finally, the update for the minimum LS error is

$$\begin{aligned} J_{\min_{k+1}} &= (\mathbf{x}-\mathbf{H}_{k+1}\hat{\boldsymbol{\theta}}_{k+1})^T(\mathbf{x}-\mathbf{H}_{k+1}\hat{\boldsymbol{\theta}}_{k+1}) \\ &= \mathbf{x}^T\mathbf{x} - \mathbf{x}^T\mathbf{H}_{k+1}\hat{\boldsymbol{\theta}}_{k+1} \\ &= \mathbf{x}^T\mathbf{x} - \mathbf{x}^T\begin{bmatrix}\mathbf{H}_k & \mathbf{h}_{k+1}\end{bmatrix}\begin{bmatrix} \hat{\boldsymbol{\theta}}_k - \dfrac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ \dfrac{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \end{bmatrix} \\ &= \mathbf{x}^T\mathbf{x} - \mathbf{x}^T\mathbf{H}_k\hat{\boldsymbol{\theta}}_k + \frac{\mathbf{x}^T\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ &\quad - \mathbf{x}^T\mathbf{h}_{k+1}\frac{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ &= J_{\min_k} - \frac{\mathbf{x}^T(\mathbf{I}-\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T)\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ &= J_{\min_k} - \frac{\mathbf{x}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{x}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ &= J_{\min_k} - \frac{\left(\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{x}\right)^2}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}}. \end{aligned}$$

Appendix 8B

Derivation of Recursive Projection Matrix

We now derive the recursive update for the projection matrix given by (8.34). Since

$$\begin{aligned}\mathbf{P}_{k+1} &= \mathbf{H}_{k+1}\left(\mathbf{H}_{k+1}^T\mathbf{H}_{k+1}\right)^{-1}\mathbf{H}_{k+1}^T \\ &= \mathbf{H}_{k+1}\mathbf{D}_{k+1}\mathbf{H}_{k+1}^T\end{aligned}$$

where $\mathbf{D}_{k+1} = (\mathbf{H}_{k+1}^T\mathbf{H}_{k+1})^{-1}$, we can use the update for $\mathbf{D}_k$ derived in Appendix 8A. As a result, we have

$$\begin{aligned}\mathbf{P}_{k+1} &= \begin{bmatrix}\mathbf{H}_k & \mathbf{h}_{k+1}\end{bmatrix}\begin{bmatrix}\mathbf{D}_k + \dfrac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} & -\dfrac{\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ -\dfrac{\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} & \dfrac{1}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}}\end{bmatrix}\begin{bmatrix}\mathbf{H}_k^T \\ \mathbf{h}_{k+1}^T\end{bmatrix} \\ &= \mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T + \frac{\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ &\quad - \frac{\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ &\quad - \frac{\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{H}_k\mathbf{D}_k\mathbf{H}_k^T}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} + \frac{\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ &= \mathbf{P}_k + \frac{\mathbf{P}_k\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{P}_k}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} - \frac{\mathbf{P}_k\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ &\quad - \frac{\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\mathbf{P}_k}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} + \frac{\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}} \\ &= \mathbf{P}_k + \frac{(\mathbf{I}-\mathbf{P}_k)\,\mathbf{h}_{k+1}\mathbf{h}_{k+1}^T\,(\mathbf{I}-\mathbf{P}_k)}{\mathbf{h}_{k+1}^T\mathbf{P}_k^\perp\mathbf{h}_{k+1}}.\end{aligned}$$

Appendix 8C

Derivation of Sequential Least Squares

In this appendix we derive (8.46)–(8.48).

$$\begin{aligned}\hat{\boldsymbol{\theta}}[n] &= \left(\mathbf{H}^T[n]\mathbf{C}^{-1}[n]\mathbf{H}[n]\right)^{-1}\mathbf{H}^T[n]\mathbf{C}^{-1}[n]\mathbf{x}[n] \\ &= \left(\begin{bmatrix}\mathbf{H}^T[n-1] & \mathbf{h}[n]\end{bmatrix}\begin{bmatrix}\mathbf{C}[n-1] & \mathbf{0} \\ \mathbf{0}^T & \sigma_n^2\end{bmatrix}^{-1}\begin{bmatrix}\mathbf{H}[n-1] \\ \mathbf{h}^T[n]\end{bmatrix}\right)^{-1} \\ &\quad \cdot\left(\begin{bmatrix}\mathbf{H}^T[n-1] & \mathbf{h}[n]\end{bmatrix}\begin{bmatrix}\mathbf{C}[n-1] & \mathbf{0} \\ \mathbf{0}^T & \sigma_n^2\end{bmatrix}^{-1}\begin{bmatrix}\mathbf{x}[n-1] \\ x[n]\end{bmatrix}\right).\end{aligned}$$

Since the covariance matrix is diagonal, it is easily inverted to yield

$$\begin{aligned}\hat{\boldsymbol{\theta}}[n] &= \left(\mathbf{H}^T[n-1]\mathbf{C}^{-1}[n-1]\mathbf{H}[n-1]+\frac{1}{\sigma_n^2}\mathbf{h}[n]\mathbf{h}^T[n]\right)^{-1} \\ &\quad \cdot\left(\mathbf{H}^T[n-1]\mathbf{C}^{-1}[n-1]\mathbf{x}[n-1]+\frac{1}{\sigma_n^2}\mathbf{h}[n]x[n]\right).\end{aligned}$$

Let

$$\boldsymbol{\Sigma}[n-1] = \left(\mathbf{H}^T[n-1]\mathbf{C}^{-1}[n-1]\mathbf{H}[n-1]\right)^{-1}$$

which is the covariance matrix of $\hat{\boldsymbol{\theta}}[n-1]$. Then,

$$\begin{aligned}\hat{\boldsymbol{\theta}}[n] &= \left(\boldsymbol{\Sigma}^{-1}[n-1]+\frac{1}{\sigma_n^2}\mathbf{h}[n]\mathbf{h}^T[n]\right)^{-1} \\ &\quad \cdot\left(\mathbf{H}^T[n-1]\mathbf{C}^{-1}[n-1]\mathbf{x}[n-1]+\frac{1}{\sigma_n^2}\mathbf{h}[n]x[n]\right).\end{aligned}$$

The first term in parentheses is just $\mathbf{\Sigma}[n]$. We can use Woodbury's identity (see Appendix 1) to obtain

$$\begin{aligned}\mathbf{\Sigma}[n] &= \left(\mathbf{\Sigma}^{-1}[n-1] + \frac{1}{\sigma_n^2}\mathbf{h}[n]\mathbf{h}^T[n]\right)^{-1} \\ &= \mathbf{\Sigma}[n-1] - \frac{\mathbf{\Sigma}[n-1]\mathbf{h}[n]\mathbf{h}^T[n]\mathbf{\Sigma}[n-1]}{\sigma_n^2 + \mathbf{h}^T[n]\mathbf{\Sigma}[n-1]\mathbf{h}[n]} \\ &= \left(\mathbf{I} - \mathbf{K}[n]\mathbf{h}^T[n]\right)\mathbf{\Sigma}[n-1]\end{aligned}$$

where

$$\mathbf{K}[n] = \frac{\mathbf{\Sigma}[n-1]\mathbf{h}[n]}{\sigma_n^2 + \mathbf{h}^T[n]\mathbf{\Sigma}[n-1]\mathbf{h}[n]}.$$

Applying these results to $\hat{\boldsymbol{\theta}}[n]$ yields

$$\begin{aligned}\hat{\boldsymbol{\theta}}[n] &= \left(\mathbf{I} - \mathbf{K}[n]\mathbf{h}^T[n]\right)\mathbf{\Sigma}[n-1] \\ &\quad \cdot \left(\mathbf{\Sigma}^{-1}[n-1]\hat{\boldsymbol{\theta}}[n-1] + \frac{1}{\sigma_n^2}\mathbf{h}[n]x[n]\right)\end{aligned}$$

since

$$\begin{aligned}\hat{\boldsymbol{\theta}}[n-1] &= \left(\mathbf{H}^T[n-1]\mathbf{C}^{-1}[n-1]\mathbf{H}[n-1]\right)^{-1}\mathbf{H}^T[n-1]\mathbf{C}^{-1}[n-1]\mathbf{x}[n-1] \\ &= \mathbf{\Sigma}[n-1]\mathbf{H}^T[n-1]\mathbf{C}^{-1}[n-1]\mathbf{x}[n-1].\end{aligned}$$

Continuing, we have

$$\begin{aligned}\hat{\boldsymbol{\theta}}[n] &= \hat{\boldsymbol{\theta}}[n-1] + \frac{1}{\sigma_n^2}\mathbf{\Sigma}[n-1]\mathbf{h}[n]x[n] - \mathbf{K}[n]\mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1] \\ &\quad - \frac{1}{\sigma_n^2}\mathbf{K}[n]\mathbf{h}^T[n]\mathbf{\Sigma}[n-1]\mathbf{h}[n]x[n].\end{aligned}$$

But

$$\begin{aligned}\frac{1}{\sigma_n^2}\mathbf{\Sigma}[n-1]\mathbf{h}[n] - \frac{1}{\sigma_n^2}\mathbf{K}[n]\mathbf{h}^T[n]\mathbf{\Sigma}[n-1]\mathbf{h}[n] \\ = \frac{1}{\sigma_n^2}\left(\sigma_n^2 + \mathbf{h}^T[n]\mathbf{\Sigma}[n-1]\mathbf{h}[n]\right)\mathbf{K}[n] \\ - \frac{1}{\sigma_n^2}\mathbf{K}[n]\mathbf{h}^T[n]\mathbf{\Sigma}[n-1]\mathbf{h}[n] \\ = \mathbf{K}[n]\end{aligned}$$

and therefore

$$\begin{aligned}\hat{\boldsymbol{\theta}}[n] &= \hat{\boldsymbol{\theta}}[n-1] + \mathbf{K}[n]x[n] - \mathbf{K}[n]\mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1] \\ &= \hat{\boldsymbol{\theta}}[n-1] + \mathbf{K}[n]\left(x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1]\right).\end{aligned}$$

Finally, to update the minimum LS error we have

$$\begin{aligned}
J_{\min}[n] &= \left(\mathbf{x}[n] - \mathbf{H}[n]\hat{\boldsymbol{\theta}}[n]\right)^T \mathbf{C}^{-1}[n]\left(\mathbf{x}[n] - \mathbf{H}[n]\hat{\boldsymbol{\theta}}[n]\right) \\
&= \mathbf{x}^T[n]\mathbf{C}^{-1}[n]\left(\mathbf{x}[n] - \mathbf{H}[n]\hat{\boldsymbol{\theta}}[n]\right) \\
&= \begin{bmatrix} \mathbf{x}^T[n-1] & x[n] \end{bmatrix} \begin{bmatrix} \mathbf{C}^{-1}[n-1] & \mathbf{0} \\ \mathbf{0}^T & \frac{1}{\sigma_n^2} \end{bmatrix} \begin{bmatrix} \mathbf{x}[n-1] - \mathbf{H}[n-1]\hat{\boldsymbol{\theta}}[n] \\ x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n] \end{bmatrix} \\
&= \mathbf{x}^T[n-1]\mathbf{C}^{-1}[n-1]\left(\mathbf{x}[n-1] - \mathbf{H}[n-1]\hat{\boldsymbol{\theta}}[n]\right) \\
&\quad + \frac{1}{\sigma_n^2}x[n]\left(x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n]\right).
\end{aligned}$$

Let $e[n] = x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1]$. Using the update for $\hat{\boldsymbol{\theta}}[n]$ produces

$$\begin{aligned}
J_{\min}[n] &= \mathbf{x}^T[n-1]\mathbf{C}^{-1}[n-1]\left(\mathbf{x}[n-1] - \mathbf{H}[n-1]\hat{\boldsymbol{\theta}}[n-1] - \mathbf{H}[n-1]\mathbf{K}[n]e[n]\right) \\
&\quad + \frac{1}{\sigma_n^2}x[n]\left(x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1] - \mathbf{h}^T[n]\mathbf{K}[n]e[n]\right) \\
&= J_{\min}[n-1] - \mathbf{x}^T[n-1]\mathbf{C}^{-1}[n-1]\mathbf{H}[n-1]\mathbf{K}[n]e[n] \\
&\quad + \frac{1}{\sigma_n^2}x[n]\left(1 - \mathbf{h}^T[n]\mathbf{K}[n]\right)e[n].
\end{aligned}$$

But

$$\mathbf{H}^T[n-1]\mathbf{C}^{-1}[n-1]\mathbf{x}[n-1] = \boldsymbol{\Sigma}^{-1}[n-1]\hat{\boldsymbol{\theta}}[n-1]$$

so that

$$\begin{aligned}
J_{\min}[n] &= J_{\min}[n-1] - \hat{\boldsymbol{\theta}}^T[n-1]\boldsymbol{\Sigma}^{-1}[n-1]\mathbf{K}[n]e[n] \\
&\quad + \frac{1}{\sigma_n^2}x[n]\left(1 - \mathbf{h}^T[n]\mathbf{K}[n]\right)e[n]
\end{aligned}$$

and also

$$\begin{aligned}
&\frac{1}{\sigma_n^2}x[n]\left(1 - \mathbf{h}^T[n]\mathbf{K}[n]\right) - \hat{\boldsymbol{\theta}}^T[n-1]\boldsymbol{\Sigma}^{-1}[n-1]\mathbf{K}[n] \\
&\quad = \frac{1}{\sigma_n^2}x[n]\left(1 - \frac{\mathbf{h}^T[n]\boldsymbol{\Sigma}[n-1]\mathbf{h}[n]}{\sigma_n^2 + \mathbf{h}^T[n]\boldsymbol{\Sigma}[n-1]\mathbf{h}[n]}\right) - \frac{\hat{\boldsymbol{\theta}}^T[n-1]\mathbf{h}[n]}{\sigma_n^2 + \mathbf{h}^T[n]\boldsymbol{\Sigma}[n-1]\mathbf{h}[n]} \\
&\quad = \frac{x[n] - \mathbf{h}^T[n]\hat{\boldsymbol{\theta}}[n-1]}{\sigma_n^2 + \mathbf{h}^T[n]\boldsymbol{\Sigma}[n-1]\mathbf{h}[n]} \\
&\quad = \frac{e[n]}{\sigma_n^2 + \mathbf{h}^T[n]\boldsymbol{\Sigma}[n-1]\mathbf{h}[n]}.
\end{aligned}$$

Hence, we have finally

$$J_{\min}[n] = J_{\min}[n-1] + \frac{e^2[n]}{\sigma_n^2 + \mathbf{h}^T[n]\boldsymbol{\Sigma}[n-1]\mathbf{h}[n]}.$$